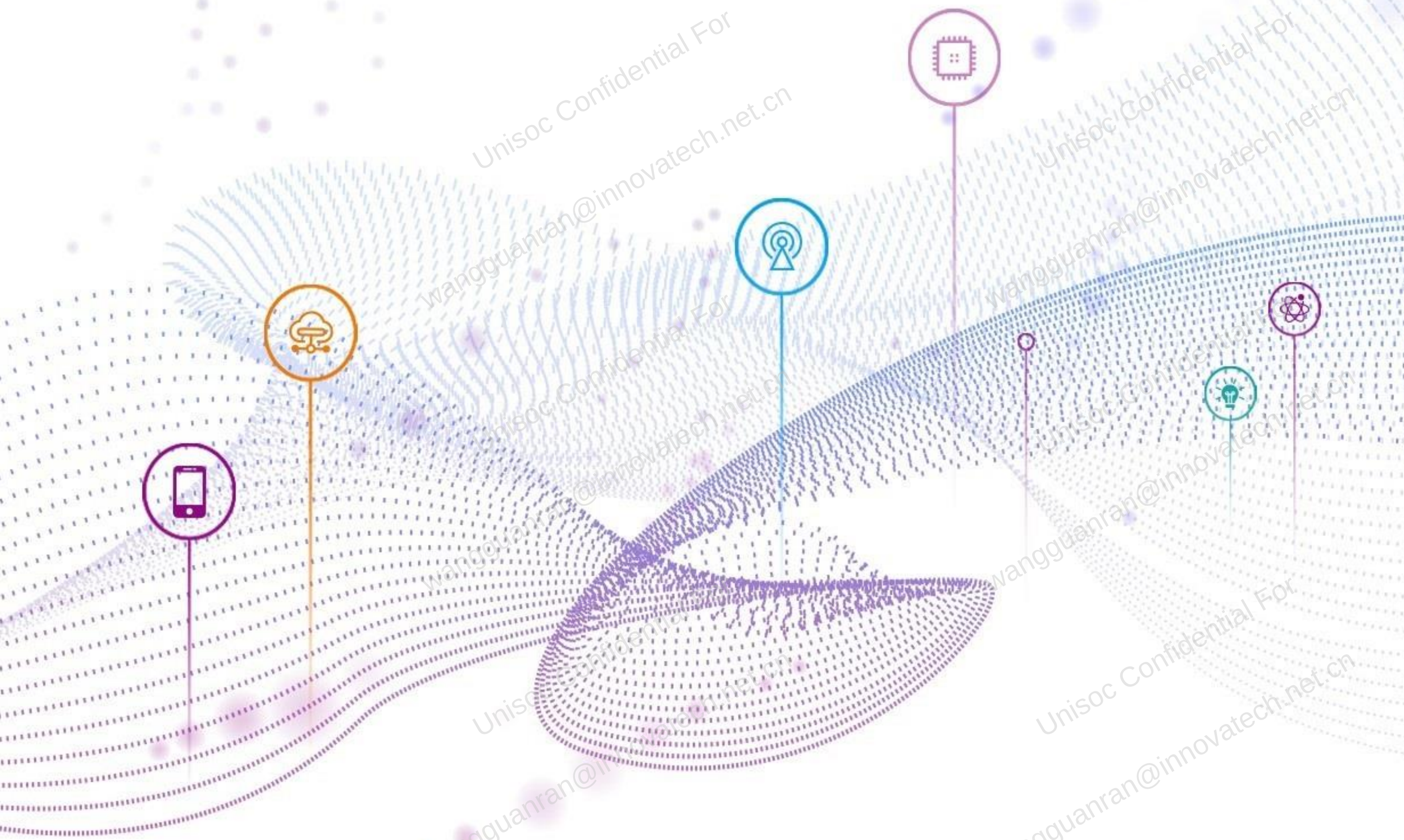

# UWS6130E Brief Device Specification

**Issue** V1.1

**Date** 2024-11-07

**UNISOC (Shanghai) Technologies Co., Ltd.**

# About This Document

## Purpose

This document describes System Overview, Package Information, Pin Information, Electrical Specification of UWS6130E.

## Intended Audience

This document is intended primarily for UWS6130E R&D, testing staff. Personnel must be familiar with the basic knowledge of UNISOC Chip.

## Acronyms and Abbreviations

| Acronym and Abbreviation | Full Name |
|---|---|
| BGA | Ball Grid Array Package |
| GPIO | General purpose input output |
| SIM | Subscriber Identity Module |
| DC | Direct Current |
| AC | Alternating Current |
| SDIO | Secure Digital Input and Output |

## Symbol Conventions

The symbols that may be found in this guide are defined in the following table.

| Symbol | Description |
|---|---|
| NOTE | Calls attention to important information, best practices and tips.<br>NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration. |
| CAUTION | Calls attention to error-prone operations.<br>CAUTION is used to address information not related to personal injury, equipment damage, and environment deterioration. |
| WARNING | Calls attention to irreversible operations.<br>WARNING is used to address information not related to personal injury and environment deterioration. |

# Change History

| Issue | Date | Description |
|---|---|---|
| V1.1 | 2024-11-07 | Update **Figure 1-1** |
| V1.0 | 2021-03-16 | This issue is the first official release. |

# Keywords

UWS6130E, Device Spec, Data Sheet.

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# 1 System Overview

## 1.1 General Description

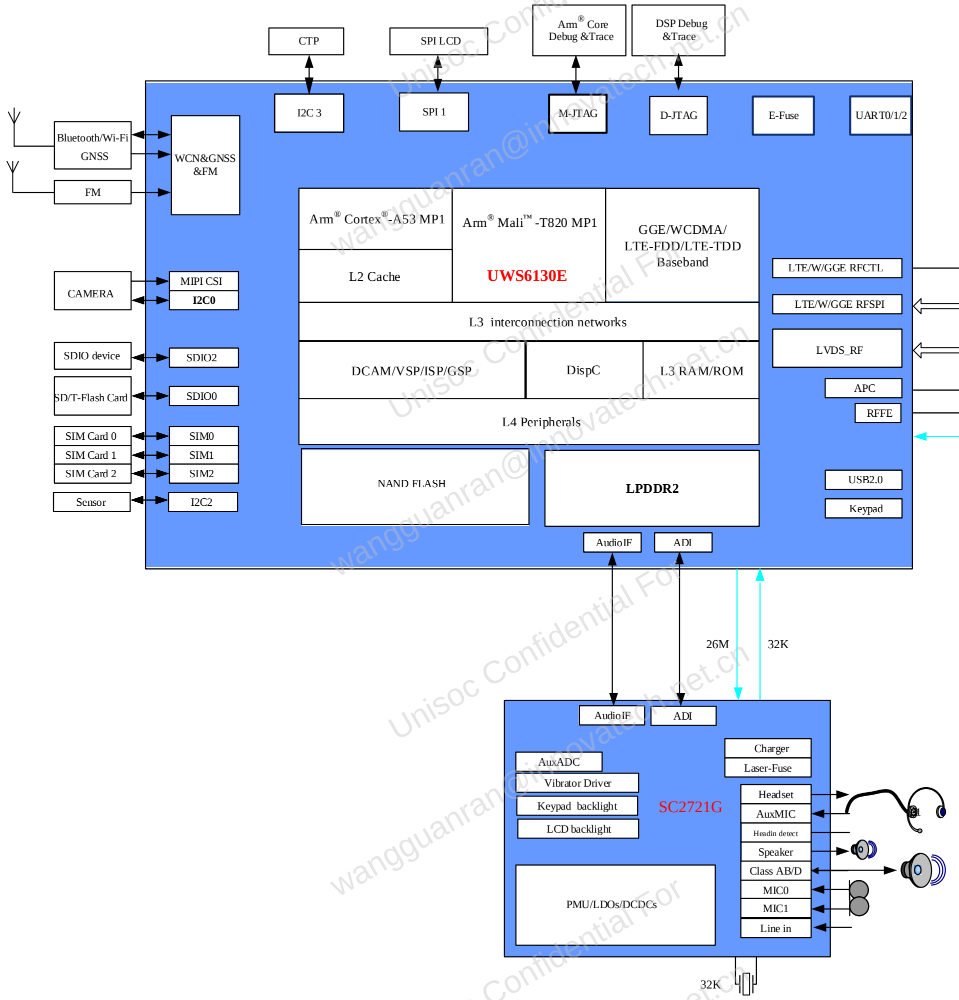
UWS6130E is a highly integrated application processor with embedded TD-LTE, LTE-FDD, WCDMA and GSM/GPRS/EDGE modem. It consists of single-core Arm® Cortex®-A53 as application processor, which includes LPDDR2 and NAND-Flash, and also integrated Wi-Fi/**Bluetooth**®/GNSS. The specially optimized architecture of UWS6130E can achieve high performance and low power for a lot of applications. Proprietary architectures and algorithms were developed for low power ASIC design and power management. Unique techniques are used for noise/offset calibration and cancellation. Overall, UWS6130E chipset presents a high cost-effective platform.

## 1.2 Chip Block Diagram

Figure 1-1 shows the chip-level functional block diagram of UWS6130E.

**Figure 1-1** UWS6130E Chip Block Diagram



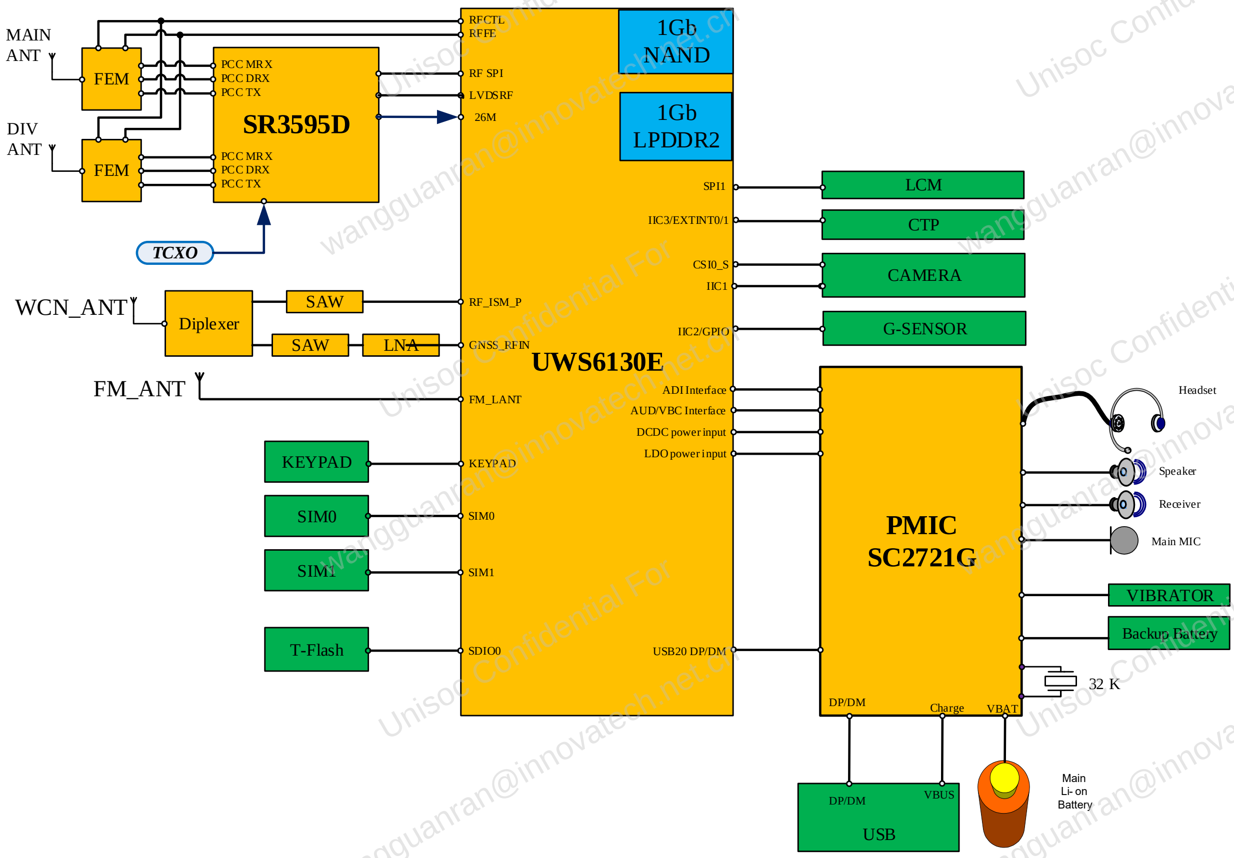

# 1.3 System Block Diagram

**Figure 1-2** shows the system block diagram of UWS6130E.



**Figure 1-2** UWS6130E System Block Diagram




# 1.4 Features

## 1.4.1 Platform Features

### AP Subsystem

- Single-core Arm® Cortex®-A53 processor, up to 1.1GHz
- 32 KB L1 I-cache and 32 KB L1 D-cache, 256KB L2 cache
- Support boot from USB, UART, SD, NAND
- Support DDR DFS technology

### Memory Interface

SIP LPDDR2 and NAND Flash

### Peripheral and Connectivity Interfaces

- Support multi-SIM cards, both 1.8V and 3.0V devices.
- Support two SDIO 3.0
- Support USB 2.0 High speed

- Support UART
- Total 3 SPI, at least one support both master and slave, support 3-wire SPI, 4-wire SPI and synchronous SPI
- Support SPI/MIPI LCD, up to VGA(MIPI) resolution
- Support 1 IIS(PCM), for audio codec connection
- Support five I2C interfaces
- Total 123 GPIO pins
- Support SIM/USIM

## 1.4.2 Modem Features

### TD-LTE/ LTE-FDD Baseband

- Compatible with 3GPP Release 12
- Support TDD/FDD LTE Category 4
- Support PWS(Public Warning System) and CMAS receiving
- Cryptographic Algorithms
  - UEA1/UIA1: SNOW-3G
  - UEA2/UIA2: AES
  - UEA3/UIA3: ZUC

### WCDMA/HSDPA/HSUPA Baseband

- Release 99 WCDMA, up to UE Class 384kbps for both uplink and downlink
- Release 7 HSDPA, up to 21Mbps(Category 14)
- Release 7 HSUPA, up to 5.76Mbps(Category 7)
- Diversity reception and soft combining over two receive antennas

### GSM/GPRS/EDGE Baseband

- Compatible with GSM/GPRS/EDGE Release 1999, GSM850, GSM900, DCS1800, and PCS1900 recommendations
- Complete in-phase and quadrature (I/Q) component interface between the DSP(Digital Signal Processor) and RF module
- EGPRS class12, type B(MCS1-9 in downlink and MCS1-9 in uplink)
- Cryptographic algorithms: A5/1 A5/2 A5/3, GEA1, GEA2, GEA3

### Multi-mode

- Support TD-LTE/GGE or TD-LTE/LTE FDD/WCDMA/GGE CSFB
- Support DSDS(L+G, L+W, L+L)
- VoLTE

## 1.4.3 Multimedia Features

### LCD Display

- Support SPI LCD, up to QVGA
- Support MIPI LCD, up to VGA

### Camera Interface

- Support up to 2M(MIPI Interface) RAW-data/YUV-data mode capture
- Support still capture while video recording with maximum resolution and displaying on local display

### Video Codec

- Support MPEG4 decoding, encoding
- Support H263 decoding (Baseline profile), encoding
- Support H264 decoding (Baseline profile), H264 encoding
- Support stream media H263/H264/MPEG4 etc. format

### Audio Codec

- Support MP3/AAC/AAC+/ AMR-NB/AMR WB/PCM/ADPCM decoding
- Audio recording codec support MP3/AMR/ACC format

## 1.4.4 WCN Features

### Wi-Fi

- Single band 2.4GHz, support 802.11b/g/n
- Support Wi-Fi and Bluetooth TDD operation and single-antenna with integrated TR-switch
- Security: WPA-TKIP, AES, WPA2
- Support 802.11w: secure management frame
- Wi-Fi/Bluetooth/LTE co-existence

### Bluetooth

- Support V2.1+EDR
- Support Bluetooth V4.2 low energy
- Bluetooth V2.1 and BLE concurrent

### GNSS

- Support GPS L1/BDS B1I/GLONASS G1
- Support GPS+BDS, GPS+GLONASS
- Support QZSS
- -160dBm Tracking sensitivity
- -147dBm Acquisition sensitivity with cold start
- TTFF: cold start<30s, hot start<2s

# 2 Package Information

## 2.1 Basic information

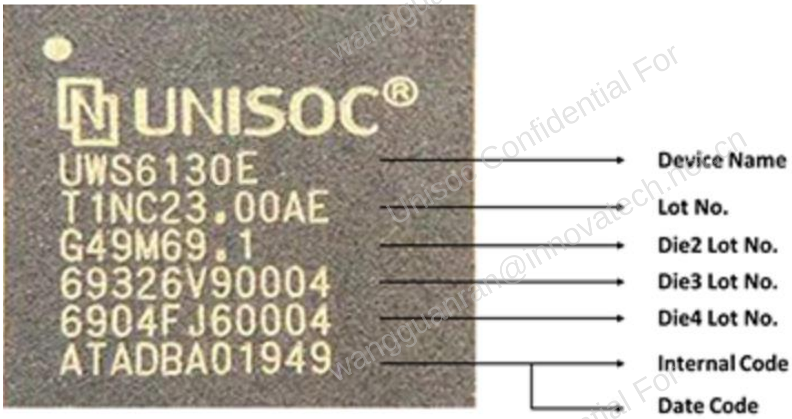
Plastic-encapsulated surface mount packages are sensitive to damage induced by absorbed moisture and temperature. UWS6130E is MSL 3, which is marked on the label of the package.

**Table 2-1** Package Information

| | |
|---|---|
| **Device Name** | UWS6130E |
| **Type** | Hybrid |
| **Body Size** | 9.4mm ×9.4mm |
| **Body Height** | 1.2mm maximum |
| **Ball count** | 444 |
| **Ball Pitch** | 0.4mm |
| **Ball Size** | 0.23mm |
| **Ball Matrix** | 23 ×23 |

## 2.2 Top Marking Definition

**Figure 2-1** Top marking definition





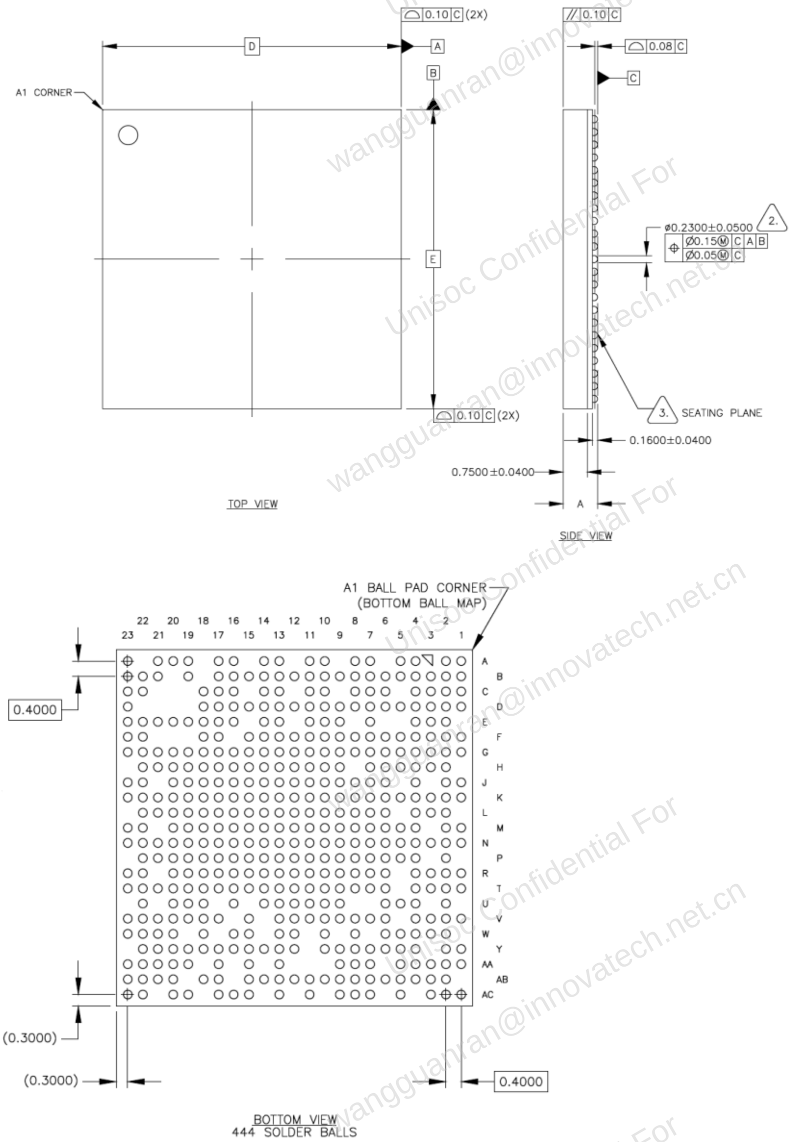
# 2.3 Ball Pinout

**Figure 2-2** Ball pinout

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | NC | SIMRST0 | | SIMRST2 | SIMCLK2 | | VDD1_LPDDR2 | RFFE0_SCK | | RFCTL6 | VREFCA | | RFSCK0 | LVDSRF0_DACON | | ZQ | VDDMEM | | FM_LANT | AVSS_FM | AVDD3V3_PA | | NC |
| B | VSIM1 | SIMDAT0 | SIMCLK0 | SIMDAT1 | SIMDAT2 | KEYIN2 | VDDMEM | RFFE0_SDA | RFCTL8 | RFCTL10 | VSS | RFCTL9 | RFSDA0 | LVDSRF0_ADCON | SPI2_CSN | SPI2_DI | TDO_LTE | | FM_SANT | | VSS | RF_ISM_P | IIS1DI |
| C | CLK26M_IN | VSIM0 | SIMCLK1 | SIMRST1 | | KEYIN0 | VSS | RFCTL3 | | GPIO9 | RFCTL11 | | RFSEN0 | SPI0_DO | | SPI2_CLK | TCK_LTE | GNSS_RF_IN | | | | WCN_CLK26M_IN | IIS1CLK |
| D | CLK26M_OUT | VSS | TSEN_VREFN | TSEN_IN | KEYOUT1 | KEYOUT2 | KEYIN1 | RFCTL7 | RFCTL4 | RFCTL5 | GPIO31 | RFCTL16 | RFCTL20 | SPI0_CLK | SPI0_DI | SPI2_DO | TDI_LTE | VSS | | | | | IIS1DO |
| E | | LVDSRF_REXT | VSS | TSEN_VREFP | | | RFCTL1 | | VSIM2 | RFCTL2 | GPIO33 | | RFCTL19 | SPI0_CSN | | TMS_ARM | AVDD1V2_GNSS | AVDD1V2_FM | VSS | VSS | VSS | AVDD1V2_TRX_ISM | IIS1LRCK |
| F | LVDSRF0_DL_PRI_P | LVDSRF0_DL_PRI_N | LVDSRF0_DL_DIV_N | LVDSRF0_DL_DIV_P | APCOUT0 | APCOUT1 | KEYOUT0 | RFCTL0 | VIO1V8 | VSS | GPIO32 | RFCTL17 | RTCK_LTE | TMS_LTE | TCK_ARM | | VSS | VSS | | | | SDA2 | SCL2 |
| G | LVDSRF0_UL_N | LVDSRF0_UL_P | VSS | VSS | VSS | VSS | AVDD1V8_BB | VSS | VDDCORE | VSS | VDDCORE | VDDCORE | VSS | VSS | VDDCORE | VIO1V8 | VSS | VSS | VSS | VSS | AVDD1V2_AFE | CLK_AUX0 | GNSS_LNA_EN |
| H | | USB_REXT | VSS | VSS | VSS | VSS | VDDCORE | | VDDCORE | VSS | VDDCORE | VDDCORE | VSS | VSS | VDDCORE | VDDCORE | VSS | VDDCORE | SDA3 | SCL3 | VIO1V8 | T_DIG | |
| J | USB_DP | USB_DM | | VSS | | VDDARM | VSS | VSS | VDDCORE | VSS | VDDCORE | VDDCORE | VSS | VSS | VDDCORE | VSS | VSS | VDDCORE | VSS | CMPD2 | | EXTINT1 | EXTINT0 |
| K | AVDD3V3_USB | VSS | AVDD1V8_USB | AVDD1V8_MPLL | CLKOUT_TEST_MPLL | VSS | VDDARM | VSS | VDDCORE | VSS | VDDCORE | VDDCORE | VSS | VSS | VDDCORE | VSS | VSS | VDDCORE | VSS | SDA0 | SCL0 | SCL1 | SDA1 |
| L | | SD2_D1 | SD2_D2 | SD2_D3 | | VSS | VDDARM | VSS | VDDARM | VSS | VDDARM | VDDCORE | VDDCORE | VSS | VDDCORE | VDDCORE | VSS | VDDCORE | CMRST2 | CMPD0 | CMRST0 | VCAM | |
| M | SD2_CLK | SD2_CMD | | SD2_D0 | VDDCORE | VSS | VDDARM | VSS | VDDARM | VSS | VDDARM | VDDCORE | VSS | VSS | VDDCORE | VSS | VSS | VDDCORE | VSS | CMRST1 | | CMMCLK0 | CMMCLK1 |
| N | VCC_NAND | VSD0 | SD0_D1 | VSD2 | VIO1V8 | VSS | VDDARM | VSS | VDDARM | VSS | VDDARM | VDDCORE | VSS | VSS | VDDCORE | VSS | VSS | VDDCORE | VSS | AVDD1V8_CSI | CMPD1 | CSI0_DN0 | CSI0_DP0 |
| P | | SD0_CLK | | SD0_D0 | SD0_CMD | VSS | VDDARM | VSS | VDDARM | VSS | VDDARM | VDDCORE | VDDCORE | VSS | VDDCORE | VDDCORE | VSS | VDDCORE | VSS | AVDD1V8_CSI | VSS | CSI0_CKN0 | |
| R | ADI_D | SD0_D3 | AUD_DAD0 | SD0_D2 | | VSS | VDDARM | VSS | VDDARM | VSS | VDDARM | VDDCORE | VSS | VSS | VDDCORE | VSS | VSS | VDDCORE | VSS | VSS | | CSI0_CKP0 | CSI0_DP1 |
| T | AUD_DASYNC | AUD_ADSYNC | ADI_SCLK | AUD_DAD1 | | VSS | VDDARM | VSS | VSS | VSS | VDDCORE | VDDCORE | VSS | VSS | VDDCORE | VSS | VSS | VDDCORE | VSS | VSS | VSS | CSI0_DN3 | CSI0_DN1 |
| U | | DCDCARM_EN | | CLK_32K | VSS | AVDD1V8_DSI | | | VEMMCIO | VDDMEM | VDDMEM | VDDMEM | VDDMEM | VDDMEM | | VDDMEM | | VDDMEM | VDDCORE | VSS | | CSI0_DP3 | |
| V | AUD_ADD0 | AUD_SCLK | EXT_RST_B | ANA_INT | | VSS | VDDCORE | VSS | VSS | VSS | VSS | VDDMEM | VSS | | VSS | | VSS | EMCA7 | VSS | VSS | VSS | CSI0_DP2 | CSI0_DN2 |
| W | | XTL_BUF_EN1 | CHIP_SLEEP | LCM_FMARK | VSS | VIO_NAND | | NF_DATA_2 | | VSS | | VDDMEM | VSS | | VSS | EMCLK_P | | EMCKE0 | | AVDD1V8_DPLL | VIO1V8 | U0RXD | U0TXD |
| Y | DSI_CKN | VSS | | LCM_RSTN | EMMC_RST | EMMC_D2 | NF_DATA_1 | NF_DATA_0 | VSS | VSS | | VSS | VSS | VSS | VSS | EMCLK_N | VSS | EMCS0_N | VSS | VSS | CLKOUT_TEST_DPLL | U0CTS | |
| AA | DSI_CKP | DSI_DN0 | VSS | EMMC_DS | EMMC_D1 | | NF_CEN0 | NF_WEN | VSS | | | | VSS | | VSS | | VSS | | VSS | EMZQ | VSS | U0RTS | U1RXD |
| AB | | DSI_DP0 | VSS | EMMC_CLK | EMMC_D5 | EMMC_D3 | EMMC_CMD | EMMC_D6 | VSS | VSS | EMDQSN1 | VDDMEM | VSS | VSS | EMVREF | | VSS | VSS | | EXTINT9 | EXTINT10 | U2TXD | U1TXD |
| AC | NC | VDDMEM | VDD1_LPDDR2 | | EMMC_D4 | | EMMC_D0 | EMMC_D7 | EMDQ8 | | EMDQS1 | | VREFDQ | | VSS | VSS | VSS | | VDDMEM | VDD1_LPDDR2 | | U2RXD | NC |

# 2.4 Package Outline

**Figure 2-3** Package outline



| SYMBOL | DIMENSION IN MM | | |
|---|---|---|---|
| | MIN | NOM | MAX |
| D | 9.3000 | 9.4000 | 9.5000 |
| E | 9.3000 | 9.4000 | 9.5000 |
| A | 1.0090 | 1.0790 | 1.1490 |

5. SOLDER BALL SIZE AFTER REFLOW IS 0.24mm.

4. REFERENCE SPECIFICATIONS:
   A. AWW SPEC #001-2234: PACKING OPERATION PROCEDURE
   B. AWW SPEC #001-2062: MARKING

3. PRIMARY DATUM C AND SEATING PLANE ARE DEFINED BY THE SPHERICAL CROWNS OF THE SOLDER BALLS

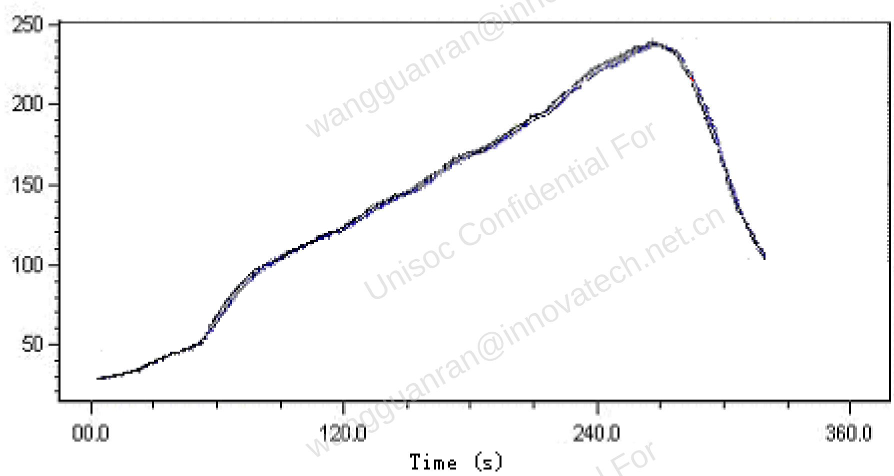
2. DIMENSION IS MEASURED AT THE MAXIMUM SOLDER BALL DIAMETER, PARALLEL TO PRIMARY DATUM C

1. ALL DIMENSIONS AND TOLERANCES CONFORM TO ASME Y14.5 – 2009

NOTES: UNLESS OTHERWISE SPECIFIED

## 2.5 Reflow Profile

**Figure 2-4** Reflow profile




- Recommended reflow profile for lead-free solder paste
  - Ramp at 1-2°C per second to 245±5°C
  - Dwell at 235°C for 10 seconds
  - Dwell at 217°C for 30~60s
  - Total reflow time is about 220~270s
  - Cold down ramp<4°C/s
- Recommended solder paste type
  - SnAgCu solder paste
  - Metal contents should be about 88.5%

- Recommended parameter for stencil making
  - Metal mask thickness: 5 mils
  - Opening area ratio: 100%

# 3 Pin Information

## 3.1 Pin Symbol Description

The following table explains the symbols used in the pin list.

**Table 3-1** Definition of pin symbols

| Field | Symbol | Type Description |
|---|---|---|
| Pin Type | I | Digital input |
| | O | Digital output |
| | O/T | Digital output with tri-state option |
| | I/O | Digital bi-directional pin |
| | I/O/T | Digital bi-directional pin with tri-state option |
| | PI | Power pin, input from external power supply |
| | PIO | Power pin, input from external or floating to use internal LDO power supply |
| | PO | Power pin, output for external devices |
| | G | Ground pin |
| | AI | Analog input pin |
| | AO | Analog output pin |
| | AIO | Analog bi-directional pin |
| Pin Value | IPU | Input with pull-up |
| | IPD | Input with pull-down |
| | OH | Output "1" |
| | OL | Output "0" |
| | Z | Tri-state |
| Power | VBAT | Battery power supply input |
| | VDDCORE | Power supply for core |
| | VDDARM | Power supply for Arm® core |
| | VIO1V8 | Power supply for I/O |
| | VDDMEM | Power supply for LPDDR IO |
| | VCAM | Power supply for Digital Camera IO |

| Field | Symbol | Type Description |
|---|---|---|
| | VSD0 | Power supply for SDIO0 IO |
| | VSD2 | Power supply for SDIO2 IO |
| | VIO_NAND | Power supply for NAND flash IO |
| | VEMMCIO | Power supply for NAND flash and EMMC IO |
| | VSIM0/VSIM1/VSIM2 | Power supply for SIM card 0/1/2 |
| | AVDD1V8_USB | Power supply for USB |
| | AVDD1V8_CSI | Power supply for CSI |
| | AVDD1V8_DSI | Power supply for DSI |
| | AVDD1V8_BB | Power supply for RF ADC/DAC and APC DAC |
| | AVDD1V8_MPLL | Power supply for Internal MPLL |
| | AVDD1V8_DPLL | Power supply for DDR PHY PLL |
| | AVDD3V3_USB | Power supply for USB |
| | AVDD3V3_PA | Power supply for Bluetooth/Wi-Fi PA |
| | AVDD1V2_AFE | Power supply for WCN Analog |
| | AVDD1V2_FM | Power supply for FM |
| | AVDD1V2_GNSS | Power supply for GNSS |
| | AVDD1V2_TRX_ISM | Power supply for WCN RF |

## 3.2 Pin List and Description

Please refer to the document *UWS6130 Pin List and Registers Instruction*.

## 3.3 Pin Multiplexed Function List

The chip adopts programmable pin multiplexing to reduce pin number as well as providing enough flexibility. Multiple signals are connected to a multiplexer that connects to the same I/O pin.

**Table 3-2** Pin multiplexed function list

| Pin Name | Function0 | Type | Function1 | Type | Function2 | Type | Function3 | Type |
|---|---|---|---|---|---|---|---|---|
| RFSDA0 | RFSDA0 | I/O/T | | | DBG_BUS20(G0) | O | GPIO1 | I/O/T |
| RFSCK0 | RFSCK0 | O/T | | | DBG_BUS21(G0) | O | GPIO2 | I/O/T |
| RFSEN0 | RFSEN0 | O/T | | | DBG_BUS22(G0) | O | GPIO3 | I/O/T |

| Pin Name | Function0 | Type | Function1 | Type | Function2 | Type | Function3 | Type |
|---|---|---|---|---|---|---|---|---|
| LVDSRF0_ADCON | RF_LVDS0_ADC_ON | O | SE_GPIO9 | I/O/T | DBG_BUS29(G0) | O | GPIO4 | I/O/T |
| LVDSRF0_DACON | RF_LVDS0_DAC_ON | O | SE_GPIO10 | I/O/T | DBG_BUS30(G0) | O | GPIO5 | I/O/T |
| RFCTL16 | RFCTL16 | O | PWMC(G1) | O | DBG_BUS15(G0) | O | GPIO7 | I/O/T |
| RFCTL17 | RFCTL17 | O | PPS(G0) | O | DBG_BUS16(G0) | O | GPIO8 | I/O/T |
| GPIO9 | | | BUA_SIM2_DET | I | DBG_BUS17(G0) | O | GPIO9 | I/O/T |
| RFCTL19 | RFCTL19 | O | U3TXD | O | DBG_BUS18(G0) | O | GPIO10 | I/O/T |
| RFCTL20 | RFCTL20 | O | U3RXD | I | DBG_BUS19(G0) | O | GPIO11 | I/O/T |
| RFCTL0 | RFCTL0 | O | | | DBG_BUS0(G0) | O | GPIO19 | I/O/T |
| RFCTL1 | RFCTL1 | O | | | DBG_BUS1(G0) | O | GPIO20 | I/O/T |
| RFCTL2 | RFCTL2 | O | | | DBG_BUS2(G0) | O | GPIO21 | I/O/T |
| RFCTL3 | RFCTL3 | O | | | DBG_BUS3(G0) | O | GPIO22 | I/O/T |
| RFCTL4 | RFCTL4 | O | | | DBG_BUS4(G0) | O | GPIO23 | I/O/T |
| RFCTL5 | RFCTL5 | O | | | DBG_BUS5(G0) | O | GPIO24 | I/O/T |
| RFCTL6 | RFCTL6 | O | | | DBG_BUS6(G0) | O | GPIO25 | I/O/T |
| RFCTL7 | RFCTL7 | O | | | DBG_BUS7(G0) | O | GPIO26 | I/O/T |
| RFCTL8 | RFCTL8 | O | | | DBG_BUS8(G0) | O | GPIO27 | I/O/T |
| RFCTL9 | RFCTL9 | O | | | DBG_BUS9(G0) | O | GPIO28 | I/O/T |
| RFCTL10 | RFCTL10 | O | | | DBG_BUS10(G0) | O | GPIO29 | I/O/T |
| RFCTL11 | RFCTL11 | O | | | DBG_BUS11(G0) | O | GPIO30 | I/O/T |
| GPIO31 | | | BUA_SIM1_DET | I | DBG_BUS12(G0) | O | GPIO31 | I/O/T |

| Pin Name | Function0 | Type | Function1 | Type | Function2 | Type | Function3 | Type |
|---|---|---|---|---|---|---|---|---|
| GPIO32 | | | PWMA | O | DBG_BUS13(G0) | O | GPIO32 | I/O/T |
| GPIO33 | | | PWMB(G1) | O | DBG_BUS14(G0) | O | GPIO33 | I/O/T |
| RFFE0_SCK | RFFE0_SCK | O | | | DBG_BUS24(G0) | O | GPIO36 | I/O/T |
| RFFE0_SDA | RFFE0_SDA | I/O/T | | | DBG_BUS25(G0) | O | GPIO37 | I/O/T |
| CMPD2 | CMPD2 | O | | | DBG_BUS18(G1) | O | GPIO40 | I/O/T |
| CMRST2 | CMRST2 | O | | | DBG_BUS19(G1) | O | GPIO41 | I/O/T |
| CMMCLK0 | CMMCLK0 | O | CLK_AUX2 | O | DBG_BUS20(G1) | O | GPIO42 | I/O/T |
| CMMCLK1 | CMMCLK1 | O | | | DBG_BUS21(G1) | O | GPIO43 | I/O/T |
| CMRST0 | CMRST0 | O | | | DBG_BUS22(G1) | O | GPIO44 | I/O/T |
| CMRST1 | CMRST1 | O | | | DBG_BUS23(G1) | O | GPIO45 | I/O/T |
| CMPD0 | CMPD0 | O | | | DBG_BUS24(G1) | O | GPIO46 | I/O/T |
| CMPD1 | CMPD1 | O | | | DBG_BUS25(G1) | O | GPIO47 | I/O/T |
| SCL0 | SCL0 | I/O/T | | | DBG_BUS26(G1) | O | GPIO48 | I/O/T |
| SDA0 | SDA0 | I/O/T | | | DBG_BUS27(G1) | O | GPIO49 | I/O/T |
| LCM_RSTN | LCM_RSTN | O | | | | | GPIO50 | I/O/T |
| LCM_FMARK | DSI_TE | I | | | | | GPIO51 | I/O/T |
| SPI2_CSN | SPI2_CSN | I/O/T | | | CM4_GPIO5 | I/O/T | GPIO52 | I/O/T |
| SPI2_DO | SPI2_DO | I/O/T | | | CM4_GPIO0 | I/O/T | GPIO53 | I/O/T |
| SPI2_DI | SPI2_DI | I/O/T | | | CM4_GPIO1 | I/O/T | GPIO54 | I/O/T |
| SPI2_CLK | SPI2_CLK | I/O/T | | | CM4_GPIO2 | I/O/T | GPIO55 | I/O/T |

| Pin Name | Function0 | Type | Function1 | Type | Function2 | Type | Function3 | Type |
|---|---|---|---|---|---|---|---|---|
| U0TXD | U0TXD | O | EXT_XTL_EN2 | I | DBG_BUS10(G1) | O | GPIO60 | I/O/T |
| U0RXD | U0RXD | I | EXT_XTL_EN3 | I | DBG_BUS11(G1) | O | GPIO61 | I/O/T |
| U0CTS | U0CTS | I | PWMC(G0) | O | DBG_BUS12(G1) | O | GPIO62 | I/O/T |
| U0RTS | U0RTS | O | SE_GPIO6 | I/O/T | DBG_BUS13(G1) | O | GPIO63 | I/O/T |
| GNSS_LNA_EN | GNSS_LNA_EN | O | | | | | GPIO69 | I/O/T |
| U1TXD | U1TXD | O | | | | | GPIO70 | I/O/T |
| U1RXD | U1RXD | I | PPS(G1) | O | | | GPIO71 | I/O/T |
| U2TXD | U2TXD | O | SE_GPIO4 | I/O/T | DBG_BUS14(G1) | O | GPIO72 | I/O/T |
| U2RXD | U2RXD | I | SE_GPIO5 | I/O/T | DBG_BUS15(G1) | O | GPIO73 | I/O/T |
| SCL1 | SCL1 | I/O/T | EXTINT14 | I | | | GPIO74 | I/O/T |
| SDA1 | SDA1 | I/O/T | EXTINT15 | I | | | GPIO75 | I/O/T |
| NF_DATA_1 | NF_DATA_1 | O | NF_DATA_1_T | I/O/T | | | GPIO76 | I/O/T |
| T_DIG | T_DIG | I | | | | | GPIO77 | I/O/T |
| EXTINT9 | EXTINT9 | I | | | BUA_TF_DET | I | GPIO78 | I/O/T |
| EXTINT10 | EXTINT10 | I | | | BAT_DET | I | GPIO79 | I/O/T |
| TCK_ARM | MTCK_ARM | I | | | | | GPIO82 | I/O/T |
| TMS_ARM | MTMS_ARM | I/O | | | | | GPIO83 | I/O/T |
| TDO_LTE | DTDO_LTE | O/T | DTDO_TWG | O/T | DBG_BUS26(G0) | O | GPIO85 | I/O/T |
| TDI_LTE | DTDI_LTE | I | DTDI_TWG | I | DBG_BUS27(G0) | O | GPIO86 | I/O/T |
| TCK_LTE | DTCK_LTE | I | DTCK_TWG | I | DBG_BUS28(G0) | O | GPIO87 | I/O/T |

| Pin Name | Function0 | Type | Function1 | Type | Function2 | Type | Function3 | Type |
|---|---|---|---|---|---|---|---|---|
| TMS_LTE | DTMS_LTE | I | DTMS_TWG | I | DBG_BUS23(G0) | O | GPIO88 | I/O/T |
| RTCK_LTE | DRTCK_LTE | O | DRTCK_TWG | O | DBG_BUS31(G0) | O | GPIO89 | I/O/T |
| SPI0_CSN | SPI0_CSN | I/O/T | | | EXTINT5 | I | GPIO90 | I/O/T |
| SPI0_DO | SPI0_DO | I/O/T | | | EXTINT6 | I | GPIO91 | I/O/T |
| SPI0_DI | SPI0_DI | I/O/T | | | EXTINT7 | I | GPIO92 | I/O/T |
| SPI0_CLK | SPI0_CLK | I/O/T | | | EXTINT8 | I | GPIO93 | I/O/T |
| EMMC_D0 | EMMC_D0 | I/O/T | NF_WPN | I/O/T | | | GPIO98 | I/O/T |
| EMMC_CMD | EMMC_CMD | I/O/T | NF_RBN | I/O/T | | | GPIO99 | I/O/T |
| EMMC_D6 | EMMC_D6 | I/O/T | NF_CLE | I/O/T | | | GPIO100 | I/O/T |
| EMMC_D7 | EMMC_D7 | I/O/T | NF_ALE | I/O/T | | | GPIO101 | I/O/T |
| EMMC_CLK | EMMC_CLK | I/O/T | NF_RE_T | I/O/T | | | GPIO102 | I/O/T |
| EMMC_D5 | EMMC_D5 | I/O/T | NF_DATA_4 | I/O/T | | | GPIO103 | I/O/T |
| EMMC_D4 | EMMC_D4 | I/O/T | NF_DATA_5 | I/O/T | | | GPIO104 | I/O/T |
| EMMC_DS | EMMC_DS | I | NF_DATA_3 | I/O/T | | | GPIO105 | I/O/T |
| EMMC_D3 | EMMC_D3 | I/O/T | NF_DATA_7 | I/O/T | | | GPIO106 | I/O/T |
| EMMC_RST | EMMC_RST | O | NF_CEN1 | I/O/T | | | GPIO107 | I/O/T |
| EMMC_D1 | EMMC_D1 | I/O/T | NF_DQS | I/O/T | | | GPIO108 | I/O/T |
| EMMC_D2 | EMMC_D2 | I/O/T | NF_DATA_6 | I/O/T | | | GPIO109 | I/O/T |
| KEYOUT0 | KEYOUT0 | O/T | EXTINT11 | I | | | GPIO121 | I/O/T |
| KEYOUT1 | KEYOUT1 | O/T | EXTINT12 | I | CM4_GPIO6 | I/O/T | GPIO122 | I/O/T |

| Pin Name | Function0 | Type | Function1 | Type | Function2 | Type | Function3 | Type |
|---|---|---|---|---|---|---|---|---|
| KEYOUT2 | KEYOUT2 | O/T | PWMB(G0) | O | CM4_GPIO7 | I/O/T | GPIO123 | I/O/T |
| KEYIN0 | KEYIN0 | I | EXTINT2 | I | DBG_BUS30(G1) | O | GPIO124 | I/O/T |
| KEYIN1 | KEYIN1 | I | EXTINT3 | I | PLL_LOCK | O | GPIO125 | I/O/T |
| KEYIN2 | KEYIN2 | I | EXTINT4 | I | DBG_BUS31(G1) | O | GPIO126 | I/O/T |
| SCL2 | SCL2 | I/O/T | | | | | GPIO127 | I/O/T |
| SDA2 | SDA2 | I/O/T | | | | | GPIO128 | I/O/T |
| CLK_AUX0 | CLK_AUX0 | O | PROBE_CLK | O | DBG_BUS29(G1) | O | GPIO129 | I/O/T |
| IIS1DI | IIS1DI | I | SE_GPIO0 | I/O/T | EXTINT13 | I | GPIO130 | I/O/T |
| IIS1DO | IIS1DO | O/T | SE_GPIO1 | I/O/T | CM4_GPIO3 | I/O/T | GPIO131 | I/O/T |
| IIS1CLK | IIS1CLK | I/O/T | SE_GPIO2 | I/O/T | CM4_GPIO4 | I/O/T | GPIO132 | I/O/T |
| IIS1LRCK | IIS1LRCK | I/O/T | SE_GPIO3 | I/O/T | EXT_XTL_EN1 | I | GPIO133 | I/O/T |
| SD2_CLK | SD2_CLK | O/T | SPI1_CLK | I/O/T | DBG_BUS0(G1) | O | GPIO134 | I/O/T |
| SD2_CMD | SD2_CMD | I/O/T | SPI1_DI | I | DBG_BUS1(G1) | O | GPIO135 | I/O/T |
| SD2_D0 | SD2_D0 | I/O/T | SPI1_DO | I/O/T | DBG_BUS2(G1) | O | GPIO136 | I/O/T |
| SD2_D1 | SD2_D1 | I/O/T | SE_GPIO7 | I/O/T | DBG_BUS3(G1) | O | GPIO137 | I/O/T |
| SD2_D2 | SD2_D2 | I/O/T | SE_GPIO8 | I/O/T | DBG_BUS4(G1) | O | GPIO138 | I/O/T |
| SD2_D3 | SD2_D3 | I/O/T | SPI1_CSN | I/O/T | DBG_BUS5(G1) | O | GPIO139 | I/O/T |
| NF_CEN0 | NF_CEN0 | O | NF_CEN0_T | I/O/T | | | GPIO140 | I/O/T |
| NF_DATA_0 | NF_DATA_0 | O | NF_DATA_0_T | I/O/T | | | GPIO141 | I/O/T |
| NF_WEN | NF_WEN | O | NF_WEN_T | I/O/T | | | GPIO142 | I/O/T |

| Pin Name | Function0 | Type | Function1 | Type | Function2 | Type | Function3 | Type |
|---|---|---|---|---|---|---|---|---|
| NF_DATA_2 | NF_DATA_2 | O | NF_DATA_2_T | I/O/T | | | GPIO143 | I/O/T |
| EXTINT0 | EXTINT0 | I | WDRST | O | SE_GPIO14 | I/O/T | GPIO144 | I/O/T |
| EXTINT1 | EXTINT1 | I | | | SE_GPIO15 | I/O/T | GPIO145 | I/O/T |
| SCL3 | SCL3 | I/O/T | | | EXT_XTL_EN0 | I | GPIO146 | I/O/T |
| SDA3 | SDA3 | I/O/T | | | | | GPIO147 | I/O/T |
| SD0_D3 | SD0_D3 | I/O/T | | | DBG_BUS6(G1) | O | GPIO148 | I/O/T |
| SD0_D2 | SD0_D2 | I/O/T | | | DBG_BUS7(G1) | O | GPIO149 | I/O/T |
| SD0_CMD | SD0_CMD | I/O/T | | | DBG_BUS8(G1) | O | GPIO150 | I/O/T |
| SD0_D0 | SD0_D0 | I/O/T | | | DBG_BUS9(G1) | O | GPIO151 | I/O/T |
| SD0_D1 | SD0_D1 | I/O/T | | | DBG_BUS28(G1) | O | GPIO152 | I/O/T |
| SD0_CLK | SD0_CLK0 | O/T | | | | | GPIO153 | I/O/T |
| SIMCLK2 | SIMCLK2 | O | SCL4 | I/O/T | SE_GPIO11 | I/O/T | GPIO154 | I/O/T |
| SIMDAT2 | SIMDA2 | I/O/T | SDA4 | I/O/T | SE_GPIO12 | I/O/T | GPIO155 | I/O/T |
| SIMRST2 | SIMRST2 | O | CLK_AUX1 | O | SE_GPIO13 | I/O/T | GPIO156 | I/O/T |
| SIMCLK0 | SIMCLK0 | O | | | | | GPIO157 | I/O/T |
| SIMDAT0 | SIMDA0 | I/O/T | | | | | GPIO158 | I/O/T |
| SIMRST0 | SIMRST0 | O | | | | | GPIO159 | I/O/T |
| SIMCLK1 | SIMCLK1 | O | | | | | GPIO160 | I/O/T |
| SIMDAT1 | SIMDA1 | I/O/T | | | | | GPIO161 | I/O/T |
| SIMRST1 | SIMRST1 | O | | | | | GPIO162 | I/O/T |

| Pin Name | Function0 | Type | Function1 | Type | Function2 | Type | Function3 | Type |
|---|---|---|---|---|---|---|---|---|
| CHIP_SLEEP | CHIP_SLEEP | O | | | | | | |
| PTEST | PTEST | I | | | | | | |
| EXT_RST_B | EXT_RST_B | I | | | | | | |
| XTL_BUF_EN1 | XTL_BUF_EN1 | O | | | | | | |
| CLK_32K | CLK_32K | I | | | | | | |
| AUD_SCLK | AUD_SCLK | O | IIS2CLK | I/O/T | | | | |
| AUD_ADD0 | AUD_ADD0 | I | IIS2DI | I | | | | |
| DCDCARM_EN | XTL_BUF_EN0 | O | | | | | | |
| ANA_INT | ANA_INT | I | | | | | | |
| AUD_ADSYNC | AUD_ADSYNC | I | | | | | | |
| ADI_SCLK | ADI_SCLK | O | | | | | | |
| AUD_DAD1 | AUD_DAD1 | O | | | | | | |
| AUD_DAD0 | AUD_DAD0 | O | IIS2DO | O/T | | | | |
| AUD_DASYNC | AUD_DASYNC | O | IIS2LRCK | I/O/T | | | | |
| ADI_D | ADI_D | I/O | | | | | | |

## 3.4 Pin status and GPIOs

**Table 3-3** Pin status and GPIOs

| Pin No. | Pin Name | GPIO | At Reset | | After Reset | | Power Domain |
|---|---|---|---|---|---|---|---|
| | | | H/L/Hiz | Pin State | H/L/Hiz | Pin State | |
| B13 | RFSDA0 | GPIO1 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| A13 | RFSCK0 | GPIO2 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| C13 | RFSEN0 | GPIO3 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |

| Pin No. | Pin Name | GPIO | At Reset | | After Reset | | Power Domain |
|---|---|---|---|---|---|---|---|
| | | | H/L/Hiz | Pin State | H/L/Hiz | Pin State | |
| B14 | LVDSRF0_ADCON | GPIO4 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| A14 | LVDSRF0_DACON | GPIO5 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| D12 | RFCTL16 | GPIO7 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| F12 | RFCTL17 | GPIO8 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| C10 | GPIO9 | GPIO9 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| E13 | RFCTL19 | GPIO10 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| D13 | RFCTL20 | GPIO11 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| F8 | RFCTL0 | GPIO19 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| E7 | RFCTL1 | GPIO20 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| E10 | RFCTL2 | GPIO21 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| C8 | RFCTL3 | GPIO22 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| D9 | RFCTL4 | GPIO23 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| D10 | RFCTL5 | GPIO24 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| A10 | RFCTL6 | GPIO25 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| D8 | RFCTL7 | GPIO26 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| B9 | RFCTL8 | GPIO27 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| B12 | RFCTL9 | GPIO28 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |

| Pin No. | Pin Name | GPIO | At Reset | | After Reset | | Power Domain |
|---|---|---|---|---|---|---|---|
| | | | H/L/Hiz | Pin State | H/L/Hiz | Pin State | |
| B10 | RFCTL10 | GPIO29 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| C11 | RFCTL11 | GPIO30 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| D11 | GPIO31 | GPIO31 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| F11 | GPIO32 | GPIO32 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| E11 | GPIO33 | GPIO33 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| A8 | RFFE0_SCK | GPIO36 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| B8 | RFFE0_SDA | GPIO37 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| J20 | CMPD2 | GPIO40 | L | INPUT (WPD) | L | INPUT (WPD) | VCAM |
| L19 | CMRST2 | GPIO41 | L | INPUT (WPD) | L | INPUT (WPD) | VCAM |
| M22 | CMMCLK0 | GPIO42 | L | OUTPUT | L | OUTPUT | VCAM |
| M23 | CMMCLK1 | GPIO43 | L | OUTPUT | L | OUTPUT | VCAM |
| L21 | CMRST0 | GPIO44 | L | INPUT (WPD) | L | INPUT (WPD) | VCAM |
| M20 | CMRST1 | GPIO45 | L | INPUT (WPD) | L | INPUT (WPD) | VCAM |
| L20 | CMPD0 | GPIO46 | L | INPUT (WPD) | L | INPUT (WPD) | VCAM |
| N21 | CMPD1 | GPIO47 | L | INPUT (WPD) | L | INPUT (WPD) | VCAM |
| K21 | SCL0 | GPIO48 | H | INPUT (WPU) | H | INPUT (WPU) | VCAM |
| K20 | SDA0 | GPIO49 | H | INPUT (WPU) | H | INPUT (WPU) | VCAM |

| Pin No. | Pin Name | GPIO | At Reset | | After Reset | | Power Domain |
|---|---|---|---|---|---|---|---|
| | | | H/L/Hiz | Pin State | H/L/Hiz | Pin State | |
| Y4 | LCM_RSTN | GPIO50 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| W4 | LCM_FMARK | GPIO51 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| B15 | SPI2_CSN | GPIO52 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| D16 | SPI2_DO | GPIO53 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| B16 | SPI2_DI | GPIO54 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| C16 | SPI2_CLK | GPIO55 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| W23 | U0TXD | GPIO60 | H | OUTPUT | H | OUTPUT | VIO1V8 |
| W22 | U0RXD | GPIO61 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| Y22 | U0CTS | GPIO62 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| AA22 | U0RTS | GPIO63 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| G23 | GNSS_LNA_EN | GPIO69 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| AB23 | U1TXD | GPIO70 | H | INPUT (WPU) | H | OUTPUT | VIO1V8 |
| AA23 | U1RXD | GPIO71 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| AB22 | U2TXD | GPIO72 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| AC22 | U2RXD | GPIO73 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| K22 | SCL1 | GPIO74 | H | INPUT (WPU) | H | INPUT (WPU) | VCAM |
| K23 | SDA1 | GPIO75 | H | INPUT (WPU) | H | INPUT (WPU) | VCAM |

| Pin No. | Pin Name | GPIO | At Reset | | After Reset | | Power Domain |
|---|---|---|---|---|---|---|---|
| | | | H/L/Hi z | Pin State | H/L/Hi z | Pin State | |
| Y7 | NF_DATA_1 | GPIO76 | H | INPUT (WPU) | H | INPUT (WPU) | VIO_NAND |
| H22 | T_DIG | GPIO77 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| AB20 | EXTINT9 | GPIO78 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| AB21 | EXTINT10 | GPIO79 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| F15 | TCK_ARM | GPIO82 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| E16 | TMS_ARM | GPIO83 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| B17 | TDO_LTE | GPIO85 | L | OUTPU T | L | OUTPU T | VIO1V8 |
| D17 | TDI_LTE | GPIO86 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| C17 | TCK_LTE | GPIO87 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| F14 | TMS_LTE | GPIO88 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| F13 | RTCK_LTE | GPIO89 | L | OUTPU T | L | OUTPU T | VIO1V8 |
| E14 | SPI0_CSN | GPIO90 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| C14 | SPI0_DO | GPIO91 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| D15 | SPI0_DI | GPIO92 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| D14 | SPI0_CLK | GPIO93 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| AC7 | EMMC_D0 | GPIO98 | H | INPUT (WPU) | H | INPUT (WPU) | VEMMCIO |
| AB7 | EMMC_CMD | GPIO99 | H | INPUT (WPU) | H | INPUT (WPU) | VEMMCIO |

| Pin No. | Pin Name | GPIO | At Reset | | After Reset | | Power Domain |
|---|---|---|---|---|---|---|---|
| | | | H/L/Hiz | Pin State | H/L/Hiz | Pin State | |
| AB8 | EMMC_D6 | GPIO100 | H | INPUT (WPU) | H | INPUT (WPU) | VEMMCIO |
| AC8 | EMMC_D7 | GPIO101 | H | INPUT (WPU) | H | INPUT (WPU) | VEMMCIO |
| AB4 | EMMC_CLK | GPIO102 | L | OUTPUT | L | OUTPUT | VEMMCIO |
| AB5 | EMMC_D5 | GPIO103 | H | INPUT (WPU) | H | INPUT (WPU) | VEMMCIO |
| AC5 | EMMC_D4 | GPIO104 | H | INPUT (WPU) | H | INPUT (WPU) | VEMMCIO |
| AA4 | EMMC_DS | GPIO105 | L | INPUT (WPD) | L | INPUT (WPD) | VEMMCIO |
| AB6 | EMMC_D3 | GPIO106 | H | INPUT (WPU) | H | INPUT (WPU) | VEMMCIO |
| Y5 | EMMC_RST | GPIO107 | H | OUTPUT | H | OUTPUT | VEMMCIO |
| AA5 | EMMC_D1 | GPIO108 | H | INPUT (WPU) | H | INPUT (WPU) | VEMMCIO |
| Y6 | EMMC_D2 | GPIO109 | H | INPUT (WPU) | H | INPUT (WPU) | VEMMCIO |
| F7 | KEYOUT0 | GPIO121 | L | OUTPUT | L | OUTPUT | VIO1V8 |
| D5 | KEYOUT1 | GPIO122 | Hiz | WPD | Hiz | WPD | VIO1V8 |
| D6 | KEYOUT2 | GPIO123 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| C6 | KEYIN0 | GPIO124 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| D7 | KEYIN1 | GPIO125 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| B6 | KEYIN2 | GPIO126 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| F23 | SCL2 | GPIO127 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |

| Pin No. | Pin Name | GPIO | At Reset | | After Reset | | Power Domain |
|---|---|---|---|---|---|---|---|
| | | | H/L/Hiz | Pin State | H/L/Hiz | Pin State | |
| F22 | SDA2 | GPIO128 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| G22 | CLK_AUX0 | GPIO129 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| B23 | IIS1DI | GPIO130 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| D23 | IIS1DO | GPIO131 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| C23 | IIS1CLK | GPIO132 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| E23 | IIS1LRCK | GPIO133 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| M1 | SD2_CLK | GPIO134 | L | INPUT (WPD) | L | INPUT (WPD) | VSD2 |
| M2 | SD2_CMD | GPIO135 | H | INPUT (WPU) | H | INPUT (WPU) | VSD2 |
| M4 | SD2_D0 | GPIO136 | H | INPUT (WPU) | H | INPUT (WPU) | VSD2 |
| L2 | SD2_D1 | GPIO137 | H | INPUT (WPU) | H | INPUT (WPU) | VSD2 |
| L3 | SD2_D2 | GPIO138 | H | INPUT (WPU) | H | INPUT (WPU) | VSD2 |
| L4 | SD2_D3 | GPIO139 | H | INPUT (WPU) | H | INPUT (WPU) | VSD2 |
| AA7 | NF_CEN0 | GPIO140 | H | INPUT (WPU) | H | INPUT (WPU) | VIO_NAND |
| Y8 | NF_DATA_0 | GPIO141 | H | INPUT (WPU) | H | INPUT (WPU) | VIO_NAND |
| AA8 | NF_WEN | GPIO142 | H | INPUT (WPU) | H | INPUT (WPU) | VIO_NAND |
| W8 | NF_DATA_2 | GPIO143 | H | INPUT (WPU) | H | INPUT (WPU) | VIO_NAND |
| J23 | EXTINT0 | GPIO144 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |

| Pin No. | Pin Name | GPIO | At Reset | | After Reset | | Power Domain |
|---|---|---|---|---|---|---|---|
| | | | H/L/Hiz | Pin State | H/L/Hiz | Pin State | |
| J22 | EXTINT1 | GPIO145 | L | INPUT (WPD) | L | INPUT (WPD) | VIO1V8 |
| H20 | SCL3 | GPIO146 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| H19 | SDA3 | GPIO147 | H | INPUT (WPU) | H | INPUT (WPU) | VIO1V8 |
| R2 | SD0_D3 | GPIO148 | L | INPUT (WPD) | L | INPUT (WPD) | VSD0 |
| R4 | SD0_D2 | GPIO149 | L | INPUT (WPD) | L | INPUT (WPD) | VSD0 |
| P5 | SD0_CMD | GPIO150 | L | INPUT (WPD) | L | INPUT (WPD) | VSD0 |
| P4 | SD0_D0 | GPIO151 | L | INPUT (WPD) | L | INPUT (WPD) | VSD0 |
| N3 | SD0_D1 | GPIO152 | L | INPUT (WPD) | L | INPUT (WPD) | VSD0 |
| P2 | SD0_CLK | GPIO153 | L | INPUT (WPD) | L | INPUT (WPD) | VSD0 |
| A5 | SIMCLK2 | GPIO154 | L | INPUT (WPD) | L | INPUT (WPD) | VSIM2 |
| B5 | SIMDAT2 | GPIO155 | L | INPUT (WPD) | L | INPUT (WPD) | VSIM2 |
| A4 | SIMRST2 | GPIO156 | L | INPUT (WPD) | L | INPUT (WPD) | VSIM2 |
| B3 | SIMCLK0 | GPIO157 | L | INPUT (WPD) | L | INPUT (WPD) | VSIM0 |
| B2 | SIMDAT0 | GPIO158 | L | INPUT (WPD) | L | INPUT (WPD) | VSIM0 |
| A2 | SIMRST0 | GPIO159 | L | INPUT (WPD) | L | INPUT (WPD) | VSIM0 |
| C3 | SIMCLK1 | GPIO160 | L | INPUT (WPD) | L | INPUT (WPD) | VSIM1 |
| B4 | SIMDAT1 | GPIO161 | L | INPUT (WPD) | L | INPUT (WPD) | VSIM1 |

| Pin No. | Pin Name | GPIO | At Reset | | After Reset | | Power Domain |
|---|---|---|---|---|---|---|---|
| | | | H/L/Hi z | Pin State | H/L/Hi z | Pin State | |
| C4 | SIMRST1 | GPIO162 | L | INPUT (WPD) | L | INPUT (WPD) | VSIM1 |

## 3.5 Pin Register Description

Please refer to the document *UWS6130 Pin List and Registers Instruction*.

## 3.6 Strapping Pins

**Table 3-4** Strapping Pins

| Ball Name | Strapping | Description | Default |
|---|---|---|---|
| ADI_SCLK | ARM_BOOT_MD2 | 000: TLC NAND ONFI<br>001: TLC NAND Toggle<br>010: MLC/SLC NAND Flash (ONFI)<br>With SIP NAND Flash in package, so we must use 010 option in SCH design.<br>011: SD Boot( no time out)<br>100: MLC/SLC NAND Flash(Toggle)<br>101: TLC NAND ONFI(Hynix)<br>110: Reserved<br>111: eMMC Boot | 1(ie=1, wpu) |
| AUD_DAD0 | ARM_BOOT_MD1 | | 1(ie=1, wpu) |
| AUD_DAD1 | ARM_BOOT_MD0 | | 1(ie=1, wpu) |
| PTEST | Chip Test | 0 :Normal<br>1: Chip Test Mode | 0 |
| U1TXD | USB Download | 0: USB download (no time out)<br>1: UART download | 1(ie=1, wpu) |
| KEYIN0 | USB Download | Press **keyin0&keyout0** or press **keyin0 to ground** when power on, system will enter USB download mode (with 16s timeout) | |
| KEYIN0+KEYIN1 | SD Boot | Press **keyin0&keyout0 + keyin1&keyout0** or press **keyin0 to ground+keyin1 to ground** when power on, system will enter SD boot mode (with timeout) | |

# 3.7 Pad Information

## 3.7.1 Digital Pad Type Description

**Table 3-5** Digital Function cell

| No. | Name | Area | Description | HV ESD |
|---|---|---|---|---|
| 1 | SPSCBC2_8X_VL/HL | 30μm×85μm | • Bi-direction Pad with Enable Controlled Pull-Down & Pull-Up Resistor.<br>• Programmable Driver Strength 2mA/4mA/6mA/8mA.<br>• No tolerance function.<br>• The Schmitt trigger input function can be control by the SE signal.<br>– SE=1, Schmitt input enable.<br>– SE=0, Schmitt input disable.<br>• Programmable soft pull up resistor 20K/4.7K. | 5kV |
| 2 | SPSEBC2_24X_VL/HL | 40μm×85μm | • Bi-direction Pad with Enable Controlled Pull-Down & Pull-Up Resistor.<br>• Programmable Driver Strength 2mA/6mA/12mA/24mA.<br>• No tolerance function.<br>• The Schmitt trigger input function can be control by the SE signal.<br>– SE=1, Schmitt input enable.<br>– SE=0, Schmitt input disable.<br>• Using for EMMC/SD interface(VDDIO=1.8V) > 100MHz. | 5kV |
| 3 | SPSCBC2_8X_IICVL/HL | 30μm×85μm | • Bi-direction Pad with Enable Controlled Pull-Down & Pull-Up Resistor.<br>• Programmable Driver Strength 2mA/4mA/6mA/8mA.<br>• No tolerance function.<br>• The Schmitt trigger input function can be control by the SE signal.<br>– SE=1, Schmitt input enable.<br>– SE=0, Schmitt input disable.<br>• Programmable soft pull up resistor 20K/1.8K. | 5kV |

| No. | Name | Area | Description | HV ESD |
|---|---|---|---|---|
| 4 | SPSCBC2_8X_W_VL/HL | 45μm×85μm | • Bi-direction Pad with Enable Controlled Pull-Down & Pull-Up Resistor.<br>• Programmable Driver Strength 2mA/4mA/6mA/8mA.<br>• No tolerance function.<br>• The Schmitt trigger input function can be control by the SE signal.<br>– SE=1, Schmitt input enable.<br>– SE=0, Schmitt input disable.<br>• For high ESD performance IO. | 8kV |

**Table 3-6** Digital Power/Ground cell

| No. | Name | Area | Description |
|---|---|---|---|
| 1 | SPVDD1ANA | 30μm×85μm | The 0.9V power supply pad, which supply the 0.9V core voltage to Core Circuit. |
| 2 | SPVDD2ANA | 30μm×85μm | The 1.8V power supply pad. Only supply the 1.8V to the internal circuit for example Efuse. Need cut the VDDIO power rail using the CUT cell. |
| 3 | SPVDD1 | 30μm×85μm | VDDCORE 0.9V power pad. |
| 4 | SPVSS1 | 30μm×85μm | Core ground pad. |
| 5 | SPVDD2 | 30μm×85μm | VDDIO 1.8V power pad. |
| 6 | SPVDD2POC | 30μm×85μm | VDDIO 1.8V power on control pad. A power domain must & only have one SPVDD2POC cell. |
| 7 | SPVSS2 | 30μm×85μm | VSSIO 1.8V ground pad. |
| 8 | SPVSS3 | 30μm×85μm | VSS/VSSIO merged ground pad. |

## 3.7.2 SPSCBC2_8X_VL/HL

SPSCBC2_8X_VL/HL is the normal pad we used usually.

**Table 3-7** Truth Table of SPSCBC2_8X_VL/HL PAD

| Function | A | PAD | C | IE | OE | WPU | WPUS | WPDO | WPDI |
|---|---|---|---|---|---|---|---|---|---|
| Input | X | 0 | 0 | 1 | 0 | 0 | X | 0 | 0 |
| | X | 1 | 1 | 1 | 0 | 0 | X | 0 | 0 |
| Output | 0 | 0 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| | 1 | 1 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| Inout | 0 | 0 | 0 | 1 | 1 | 0 | X | 0 | 0 |

| Function | A | PAD | C | IE | OE | WPU | WPUS | WPDO | WPDI |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1 | 1 | 1 | 1 | 0 | X | 0 | 0 |
| Tri State | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| Pull up | X | 1 | 0 | 0 | 0 | 1 | 0: 20K(1.8V) | 0 | 0 |
| | | | | | | | 1: 4.7K(1.8V) | | |
| Pull Down | X | 0 | 0 | 0 | 0 | 0 | X | 1 | 0 |
| IO Power-down | X | X | 1 | X | X | X | X | X | 1 |

Input function C=PAD&IE, Output function PAD=A&OE

**Table 3-8** Drive Strength Select Function of SPSCBC2_8X_VL/HL PAD

| DRV[1] | DRV[0] | Drive Strength |
|---|---|---|
| 0 | 0 | 2mA |
| 0 | 1 | 4mA |
| 1 | 0 | 6mA |
| 1 | 1 | 8mA |

All Corner (SS)@0.2 × VDDIO; 0.8Xvddio

**Table 3-9** DC Parameter of SPSCBC2_8X_VL/HL PAD

| Parameter | Parameter Description | SS | TT | FF |
|---|---|---|---|---|
| $V_{T}+$ | Schmitt trig. Low to High threshold point (1.8V) | X | >1.0V | X |
| $V_{T}-$ | Schmitt trig. High to low threshold point (1.8V) | X | >0.8V | X |
| RPU | Pull-up resistor(1.8V) | 20K/4.7K ±50% | 20K/4.7K ±50% | 20K/4.7K ±50% |
| RPD | Pull-down resistor(1.8V) | 50K ±70% | 50K ±70% | 50K ±70% |
| Ist | Standby Current | <2µA | <2µA | <2µA |
| IoZ | Tri-state output leakage current@Vo=3V or 0V | <2µA | <2µA | <2µA |

**Table 3-10** AC Parameter of SPSCBC2_8X_VL/HL PAD

| Delay Path | | Rise Time(ns) | Fall Time(ns) | Rise Delay(ns) | Fall Delay(ns) |
|---|---|---|---|---|---|
| PAD->C | | 0.4771 | 0.959 | 0.5199 | 1.079 |
| A-PAD | 2mA | 7.931 | 7.907 | 13.45 | 12.577 |

| Delay Path | | Rise Time(ns) | Fall Time(ns) | Rise Delay(ns) | Fall Delay(ns) |
|---|---|---|---|---|---|
| | 4mA | 4.469 | 4.326 | 5.415 | 5.08 |
| | 6mA | 3.545 | 3.368 | 3.327 | 3.128 |
| | 8mA | 3.171 | 2.961 | 2.487 | 2.316 |

The data simulation is SS corner, $C_{loading}$=0.3pF, $C_{pad}$=30pF, VDDIO=1.8V, Temp=125℃.

### 3.7.3 SPSEBC2_24X_VL/HL

SPSEBC2_24X_VL/HL IO PAD is design for eMMC. The function is a managed memory capable of storing code and data. It is specifically designed for mobile devices. The eMMC is intended to offer the performance and features required by mobile devices while maintaining low power consumption. The eMMC device contains features that support high throughput for large data transfers and performance for small random data more commonly found in code usage. It also contains many security features.

**Table 3-11** Truth Table of SPSEBC2_24X_VL/HL PAD

| Function | A | PAD | C | IE | OE | WPU | WPUS | WPDO | WPDI |
|---|---|---|---|---|---|---|---|---|---|
| Input | X | 0 | 0 | 1 | 0 | 0 | X | 0 | 0 |
| | X | 1 | 1 | 1 | 0 | 0 | X | 0 | 0 |
| Output | 0 | 0 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| | 1 | 1 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| Inout | 0 | 0 | 0 | 1 | 1 | 0 | X | 0 | 0 |
| | 1 | 1 | 1 | 1 | 1 | 0 | X | 0 | 0 |
| Tri State | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| Pull up | X | 1 | 0 | 0 | 0 | 1 | 0: 20K(1.8V) | 0 | 0 |
| | | | | | | | 1: 4.7K(1.8V) | | |
| Pull Down | X | 0 | 0 | 0 | 0 | 0 | X | 1 | 0 |
| IO Power-down | X | X | 1 | X | X | X | X | X | 1 |

Input function C=PAD&IE, Output function PAD=A&OE

**Table 3-12** Drive Strength Select Function of SPSEBC2_24X_VL/HL PAD

| DRV[2] | DRV[1] | DRV[0] | Drive Strength | Note |
|---|---|---|---|---|
| 0 | 0 | 0 | 200Ω | Option |
| 0 | 0 | 1 | 100Ω | Option |
| 0 | 1 | 0 | 70Ω | Option |
| 0 | 1 | 1 | 50Ω | Mandatory |

| DRV[2] | DRV[1] | DRV[0] | Drive Strength | Note |
|---|---|---|---|---|
| 1 | 0 | 0 | 40Ω | Option |
| 1 | 0 | 1 | 33Ω | Option |
| 1 | 1 | 0 | 28Ω | Option |
| 1 | 1 | 1 | 25Ω | Option |

All Corner (TT)@0.5 × VDDIO

**Table 3-13** DC Parameter of SPSEBC2_24X_VL/HL PAD

| Parameter | Parameter Description | SS | TT | FF |
|---|---|---|---|---|
| VT+ | Schmitt trig. Low to High threshold point (1.8V) | X | >1.0V | X |
| VT- | Schmitt trig. High to low threshold point (1.8V) | X | >0.8V | X |
| RPU | Pull-up resistor(1.8V) | 20K/4.7K±50% | 20K/4.7K±50% | 20K/4.7K±50% |
| RPD | Pull-down resistor(1.8V) | 50K±70% | 50K±70% | 50K±70% |
| Ist | Standby Current | <2μA | <2μA | <2μA |
| IoZ | Tri-state output leakage current@Vo=1.8V or 0V | <2μA | <2μA | <2μA |

**Table 3-14** AC Parameter of SPSEBC2_24X_VL/HL PAD

| Delay Path | | Rise Time(ns) | Fall Time(ns) | Rise Delay(ns) | Fall Delay(ns) |
|---|---|---|---|---|---|
| PAD->C | | 0.51 | 0.736 | 0.162 | 0.153 |
| A-PAD | 200Ω | 5.594 | 14.547 | 9.386 | 25.06 |
| | 100Ω | 3.031 | 6.634 | 3.774 | 10.637 |
| | 70Ω | 2.365 | 4.457 | 2.315 | 6.246 |
| | 50Ω | 1.976 | 3.443 | 1.666 | 4.31 |
| | 40Ω | 2.029 | 2.956 | 1.349 | 3.275 |
| | 35Ω | 1.83 | 2.57 | 1.131 | 2.631 |
| | 30Ω | 1.728 | 2.311 | 0.969 | 2.198 |
| | 25Ω | 1.615 | 2.111 | 0.858 | 1.886 |

NOTE

The data simulation is SS corner, clk=208MHz, $C_{loading}$=0.2pF VDDIO=1.8V, Temp=125°C.

### 3.7.4 SPSCBC2_8X_IIC_VL/HL

**Table 3-15** Truth Table of SPSCBC2_8X_IIC_VL/HL PAD

| Function | A | PAD | C | IE | OE | WPU | WPUS | WPDO | WPDI |
|---|---|---|---|---|---|---|---|---|---|
| Input | X | 0 | 0 | 1 | 0 | 0 | X | 0 | 0 |
| | X | 1 | 1 | 1 | 0 | 0 | X | 0 | 0 |
| Output | 0 | 0 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| | 1 | 1 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| Inout | 0 | 0 | 0 | 1 | 1 | 0 | X | 0 | 0 |
| | 1 | 1 | 1 | 1 | 1 | 0 | X | 0 | 0 |
| Tri State | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| Pull up | X | 1 | 0 | 0 | 0 | 1 | 0: 20K(1.8V) | 0 | 0 |
| | | | | | | | 1: 1.8K(1.8V) | | |
| Pull Down | X | 0 | 0 | 0 | 0 | 0 | X | 1 | 0 |
| IO Power-down | X | X | 1 | X | X | X | X | X | 1 |

Input function C=PAD&IE, Output function PAD=A&OE

**Table 3-16** Drive Strength Select Function of SPSCBC2_8X_IIC_VL/HL PAD

| DRV[1] | DRV[0] | Drive Strength |
|---|---|---|
| 0 | 0 | 2mA |
| 0 | 1 | 4mA |
| 1 | 0 | 6mA |
| 1 | 1 | 8mA |

All Corner (SS)@0.2 × VDDIO; 0.8×VDDIO

**Table 3-17** DC Parameter of SPSCBC2_8X_IIC_VL/HL PAD

| Parameter | Parameter Description | SS | TT | FF |
|---|---|---|---|---|
| $V_T$+ | Schmitt trig. Low to High threshold point (1.8V) | X | >1.0V | X |
| $V_T$- | Schmitt trig. High to low threshold point (1.8V) | X | >0.8V | X |
| RPU | Pull-up resistor(1.8V) | 20K/1.8K±50% | 20K/1.8K±50% | 20K/1.8K±50% |
| RPD | Pull-down resistor(1.8V) | 50K±70% | 50K±70% | 50K±70% |
| Ist | Standby Current | <2µA | <2µA | <2µA |

| Parameter | Parameter Description | SS | TT | FF |
|---|---|---|---|---|
| IoZ | Tri-state output leakage current@Vo=3V or 0V | <2µA | <2µA | <2µA |

**Table 3-18** AC Parameter of SPSCBC2_8X_IIC_VL/HL PAD

| Delay Path | | Rise Time(ns) | Fall Time(ns) | Rise Delay(ns) | Fall Delay(ns) |
|---|---|---|---|---|---|
| PAD->C | | 0.4771 | 0.959 | 0.5199 | 1.079 |
| A-PAD | 2mA | 7.931 | 7.907 | 13.45 | 12.577 |
| | 4mA | 4.469 | 4.326 | 5.415 | 5.08 |
| | 6mA | 3.545 | 3.368 | 3.327 | 3.128 |
| | 8mA | 3.171 | 2.961 | 2.487 | 2.316 |

The data simulation is SS corner, $C_{loading}$=0.3pF, $C_{pad}$=30pF, VDDIO=1.8V, Temp=125℃.

### 3.7.5 SPSCBC2_8X_W_VL/HL

Base on the SPSCBC2_8X_VL/HL, enlarge the ESD area for Keypad and etc 8kV ESD performance.

**Table 3-19** Truth Table of SPSCBC2_8X_W_VL/HL PAD

| Function | A | PAD | C | IE | OE | WPU | WPUS | WPDO | WPDI |
|---|---|---|---|---|---|---|---|---|---|
| Input | X | 0 | 0 | 1 | 0 | 0 | X | 0 | 0 |
| | X | 1 | 1 | 1 | 0 | 0 | X | 0 | 0 |
| Output | 0 | 0 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| | 1 | 1 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| Inout | 0 | 0 | 0 | 1 | 1 | 0 | X | 0 | 0 |
| | 1 | 1 | 1 | 1 | 1 | 0 | X | 0 | 0 |
| Tri State | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| Pull up | X | 1 | 0 | 0 | 0 | 1 | 0: 20K(1.8V) | 0 | 0 |
| | | | | | | | 1: 4.7K(1.8V) | | |
| Pull Down | X | 0 | 0 | 0 | 0 | 0 | X | 1 | 0 |
| IO Power-down | X | X | 1 | X | X | X | X | X | 1 |

Input function C=PAD&IE, Output function PAD=A&OE

**Table 3-20** Drive Strength Select Function of SPSCBC2_8X_W_VL/HL PAD

| DRV[1] | DRV[0] | Drive Strength |
|---|---|---|
| 0 | 0 | 2mA |
| 0 | 1 | 4mA |
| 1 | 0 | 6mA |
| 1 | 1 | 8mA |

All Corner (SS)@0.2 × VDDIO; 0.8×VDDIO

**Table 3-21** DC Parameter of SPSCBC2_8X_W_VL/HL PAD

| Parameter | Parameter Description | SS | TT | FF |
|---|---|---|---|---|
| $V_T$+ | Schmitt trig. Low to High threshold point (1.8V) | X | >1.0V | X |
| $V_T$- | Schmitt trig. High to low threshold point (1.8V) | X | >0.8V | X |
| RPU | Pull-up resistor(1.8V) | 20K/4.7K±50% | 20K/4.7K±50% | 20K/4.7K±50% |
| RPD | Pull-down resistor(1.8V) | 50K±70% | 50K±70% | 50K±70% |
| Ist | Standby Current | <2μA | <2μA | <2μA |
| IoZ | Tri-state output leakage current@Vo=3V or 0V | <2μA | <2μA | <2μA |

**Table 3-22** AC Parameter of SPSCBC2_8X_W_VL/HL PAD

| Delay Path | | Rise Time(ns) | Fall Time(ns) | Rise Delay(ns) | Fall Delay(ns) |
|---|---|---|---|---|---|
| PAD->C | | 0.57 | 0.49 | 0.162 | 0.153 |
| A-PAD | 2mA | 8.092 | 8.08 | 13.701 | 12.797 |
| | 4mA | 4.543 | 4.407 | 5.538 | 5.197 |
| | 6mA | 3.593 | 3.42 | 3.405 | 3.201 |
| | 8mA | 3.208 | 3.001 | 2.54 | 2.637 |

The data simulation is SS corner, $C_{loading}$=0.3pF, $C_{pad}$=30pF, VDDIO=1.8V, Temp=125°C.

## 3.7.6 SPSCBC2_8X_W_IIC_VL/HL

Base on the SPSCBC2_8X_W_VL/HL, change the soft pull up resistor from 4.7K to 1.8K. HBM 8kV ESD performance.

**Table 3-23** Truth Table of SPSCBC2_8X_W_IIC_VL/HL PAD

| Function | A | PAD | C | IE | OE | WPU | WPUS | WPDO | WPDI |
|---|---|---|---|---|---|---|---|---|---|
| Input | X | 0 | 0 | 1 | 0 | 0 | X | 0 | 0 |

| Function | A | PAD | C | IE | OE | WPU | WPUS | WPDO | WPDI |
|---|---|---|---|---|---|---|---|---|---|
| | X | 1 | 1 | 1 | 0 | 0 | X | 0 | 0 |
| Output | 0 | 0 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| | 1 | 1 | 0 | 0 | 1 | 0 | X | 0 | 0 |
| Inout | 0 | 0 | 0 | 1 | 1 | 0 | X | 0 | 0 |
| | 1 | 1 | 1 | 1 | 1 | 0 | X | 0 | 0 |
| Tri State | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| | X | HZ | 0 | 0 | 0 | 0 | X | 0 | 0 |
| Pull up | X | 1 | 0 | 0 | 0 | 1 | 0: 20K(1.8V) | 0 | 0 |
| | | | | | | | 1: 1.8K(1.8V) | | |
| Pull Down | X | 0 | 0 | 0 | 0 | 0 | X | 1 | 0 |
| IO Power-down | X | X | 1 | X | X | X | X | X | 1 |

Input function C=PAD&IE, Output function PAD=A&OE

**Table 3-24** Drive Strength Select Function of SPSCBC2_8X_W_IIC_VL/HL PAD

| DRV[1] | DRV[0] | Drive Strength |
|---|---|---|
| 0 | 0 | 2mA |
| 0 | 1 | 4mA |
| 1 | 0 | 6mA |
| 1 | 1 | 8mA |

All Corner (SS)@0.2 × VDDIO; 0.8×VDDIO

**Table 3-25** DC Parameter of SPSCBC2_8X_W_IIC_VL/HL PAD

| Parameter | Parameter Description | SS | TT | FF |
|---|---|---|---|---|
| VT+ | Schmitt trig. Low to High threshold point (1.2V) | X | 0.80V | X |
| VT- | Schmitt trig. High to low threshold point (1.2V) | X | 0.39V | X |
| VT+ | Schmitt trig. Low to High threshold point (1.8V) | X | 1.20V | X |
| VT- | Schmitt trig. High to low threshold point (1.8V) | X | 0.64V | X |
| RPU | Pull-up resistor(1.8V) | 30K/1.1K | 22K/1.8K | 16K/2.6K |
| RPD | Pull-down resistor(1.8V) | 30K | 40K | 60K |

| Parameter | Parameter Description | SS | TT | FF |
|---|---|---|---|---|
| Iavg | Active Current@100MHz, FF Corner, 30pF loading | X | X | 4mA |
| Ist | Standby Current | X | X | <0.1μA |
| IoZ | Tri-state output leakage current@Vo=3V or 0V | X | X | <0.1μA |

**Table 3-26** AC Parameter of SPSCBC2_8X_W_IIC_VL/HL PAD

| Delay Path | | Rise Time(ns) | Fall Time(ns) | Rise Delay(ns) | Fall Delay(ns) |
|---|---|---|---|---|---|
| PAD->C | | 0.193 | 0.198 | 0.437 | 0.327 |
| A-PAD | 2mA | 4.911 | 4.766 | 3.379 | 3.54 |
| | 4mA | 2.569 | 2.491 | 2.452 | 2.51 |
| | 6mA | 1.829 | 1.778 | 2.168 | 2.193 |
| | 8mA | 2.609 | 2.548 | 2.491 | 2.521 |

The data simulation is SS corner, $C_{loading}$=0.3pF, $C_{pad}$=30pF/15pF, VDDIO=1.8V, Temp=125°C.

## 3.7.7 SPPDWUWSWCDG_V/H

**Figure 3-1** SPPDWUWSWCDG_V/H PAD

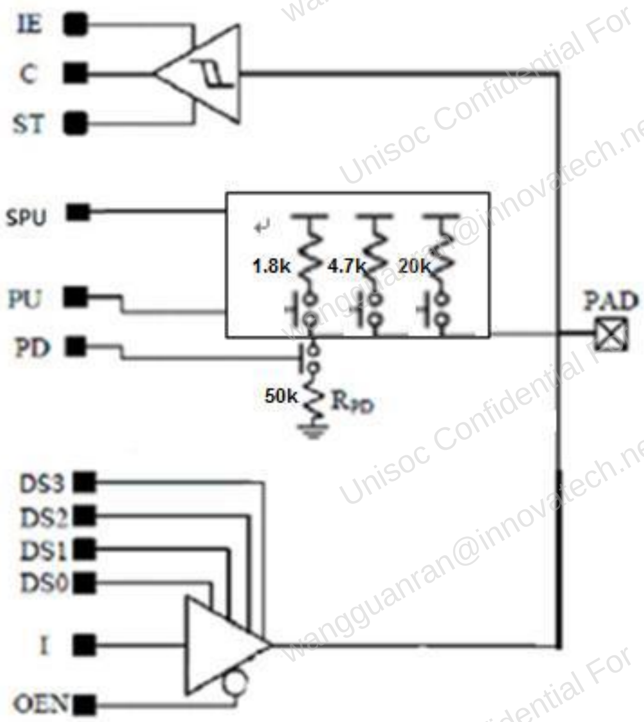


**Table 3-27** Truth Table of SPPDWUWSWCDG_V/H PAD

| INPUT | | | | | | | | | | | | | OUTPUT | | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MS | OEN | I | DS0 | DS1 | DS2 | DS3 | PU | SPU | PD | ST | IE | PAD | PAD | C | |
| 1 | 0 | 0 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | - | 0 | 0 | 1.8V UD operation |
| 1 | 0 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | - | 1 | 0 | |
| 1 | 0 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | - | 1 | 1 | |
| 0 | 0 | 0 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | - | 0 | 0 | 3.0V OD operation |
| 0 | 0 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | - | 1 | 0 | |
| 0 | 0 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | - | 1 | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | - | 0 | Input mode |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 1 | - | 0 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 1 | - | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 0 | 0 | 0/1 | 0 | Z | - | 0 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 0 | 0 | 0/1 | 1 | Z | - | X | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 0 | 1 | 0/1 | 0/1 | N.A. | - | N.A. | Forbidden |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 1 | 1 | 0/1 | 0/1 | | - | | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 0 | 0 | 0/1 | 0 | Z | H | 0 | 20k weak pull |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 0 | 0 | 0/1 | 1 | Z | H | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 1 | 0 | 0/1 | 0 | Z | H | 0 | 4.7k strong pull |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 1 | 0 | 0/1 | 1 | Z | H | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 1 | 0 | 0/1 | 0 | Z | H | 0 | 1.8k strong pull |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 1 | 0 | 0/1 | 1 | Z | H | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 0 | 1 | 0/1 | 0/1 | Z | L | 0 | Pull down |

Input function C=PAD&IE, Output function PAD=A&OE

**Table 3-28** Drive Strength Select Function of SPPDWUWSWCDG_V/H PAD

| DS3 | DS2 | DS1 | DS0 | Drive Strength |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 2.4mA |
| 0 | 0 | 0 | 1 | 5.5mA |
| 0 | 0 | 1 | 0 | 7.5mA |
| 0 | 0 | 1 | 1 | 10.4mA |
| 0 | 1 | 0 | 0 | 12.4mA |
| 0 | 1 | 0 | 1 | 15mA |
| 0 | 1 | 1 | 0 | 17.4mA |
| 0 | 1 | 1 | 1 | 20mA |
| 1 | 0 | 0 | 0 | 24.6mA |
| 1 | 0 | 0 | 1 | 27.8mA |
| 1 | 0 | 1 | 0 | 29.6mA |
| 1 | 0 | 1 | 1 | 32mA |
| 1 | 1 | 0 | 0 | 34.2mA |
| 1 | 1 | 0 | 1 | 37mA |
| 1 | 1 | 1 | 0 | 39mA |
| 1 | 1 | 1 | 1 | 42mA |

All Corner (SS)@0.5 × VDDIO

### 3.7.8 SPPDWUWSWSIM_V/H

**Figure 3-2** SPPDWUWSWSIM_V/H PAD

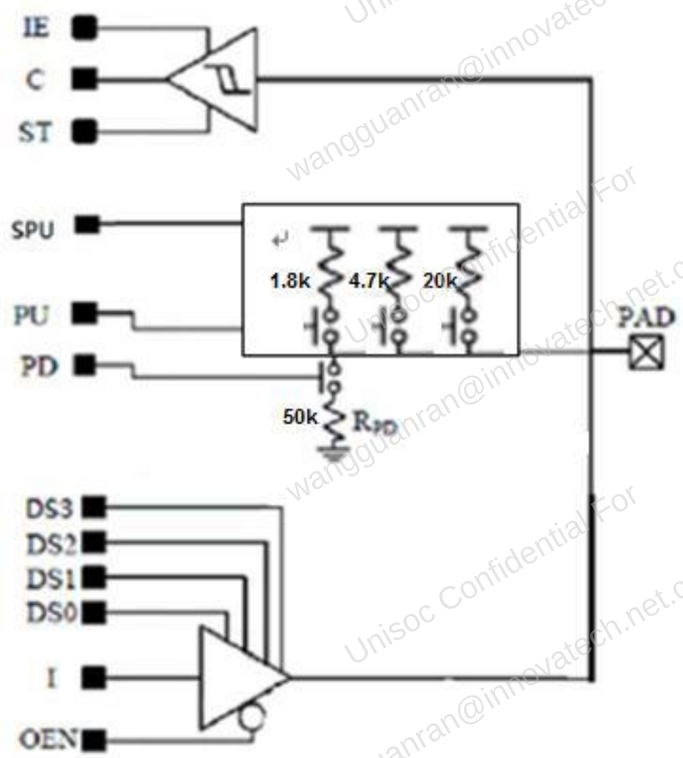



**Table 3-29** Truth Table of SPPDWUWSWSIM_V/H PAD

| INPUT | | | | | | | | | | | | | OUTPUT | | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MS | OEN | I | DS0 | DS1 | DS2 | DS3 | PU | SPU | PD | ST | IE | PAD | PAD | C | |
| 1 | 0 | 0 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | - | 0 | 0 | 1.8V UD operation |
| 1 | 0 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | - | 1 | 0 | |
| 1 | 0 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | - | 1 | 1 | |
| 0 | 0 | 0 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | - | 0 | 0 | 3.0V OD operation |
| 0 | 0 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | - | 1 | 0 | |
| 0 | 0 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | - | 1 | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | - | 0 | Input mode |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 1 | - | 0 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 1 | - | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 0 | 0 | 0/1 | 0 | Z | - | 0 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 0 | 0 | 0/1 | 1 | Z | - | X | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 0 | 1 | 0/1 | 0/1 | N.A. | - | N.A. | Forbidden |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 1 | 1 | 0/1 | 0/1 | | - | | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 0 | 0 | 0/1 | 0 | Z | H | 0 | 20k weak pull |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 0 | 0 | 0/1 | 1 | Z | H | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 1 | 0 | 0/1 | 0 | Z | H | 0 | 4.7k strong pull |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 1 | 0 | 0/1 | 1 | Z | H | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 1 | 0 | 0/1 | 0 | Z | H | 0 | 1.8k strong pull |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 1 | 1 | 0 | 0/1 | 1 | Z | H | 1 | |
| 0/1 | 1 | 0/1 | 0/1 | 0/1 | 0/1 | 0/1 | 0 | 0 | 1 | 0/1 | 0/1 | Z | L | 0 | Pull down |

Input function C=PAD&IE, Output function PAD=A&OE



**Table 3-30** Drive Strength Select Function of SPPDWUWSWSIM_V/H PAD

| DS3 | DS2 | DS1 | DS0 | Drive Strength |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 2.4mA |
| 0 | 0 | 0 | 1 | 5.5mA |
| 0 | 0 | 1 | 0 | 7.5mA |
| 0 | 0 | 1 | 1 | 10.4mA |
| 0 | 1 | 0 | 0 | 12.4mA |
| 0 | 1 | 0 | 1 | 15mA |
| 0 | 1 | 1 | 0 | 17.4mA |
| 0 | 1 | 1 | 1 | 20mA |
| 1 | 0 | 0 | 0 | 24.6mA |
| 1 | 0 | 0 | 1 | 27.8mA |
| 1 | 0 | 1 | 0 | 29.6mA |
| 1 | 0 | 1 | 1 | 32mA |
| 1 | 1 | 0 | 0 | 34.2mA |
| 1 | 1 | 0 | 1 | 37mA |
| 1 | 1 | 1 | 0 | 39mA |
| 1 | 1 | 1 | 1 | 42mA |

All Corner (SS)@0.5 × VDDIO



# 4 Electrical Specification

## 4.1 DC specification

### 4.1.1 Absolute Maximum Ratings

The functionality is subject to the absolute maximum/minimum values listed in **Table 4-1**. Do not exceed these parameters or the part may be damaged permanently. Operation at absolute maximum ratings is not guaranteed.

**Table 4-1** Absolute maximum ratings

| Symbol | Parameter | Min | Max | Unit |
|---|---|---|---|---|
| $V_{DI\_18V}$ | Input voltage on 1.8V domain digital input | -0.3 | 1.98 | V |
| $V_{DI\_3V}$ | Input voltage on 3V domain digital input | -0.3 | 3.3 | V |
| $V_{AI\_18V}$ | Input voltage on 1.8V domain analog input | -0.3 | 1.98 | V |
| $V_{max, ESD}$ | Maximum ESD stress voltage, Human Body Model, any pin to any supply pin, either polarity or any pin to all non-supply pins together, either polarity. Three stresses maximum. | | 2,000 | V |
| $I_{max, DC}$ | Maximum DC Input current for any non-supply pin | | 5 | mA |
| $T_{storage}$ | Storage temperature | -40 | +125 | ℃ |
| $T_c$ | Device operating temperature | -20 | +60 | ℃ |
| | Fuse programming temperature | -20 | +60 | ℃ |
| TJMAX | Max junction temperature | | +125 | ℃ |

### 4.1.2 Recommended Operating Conditions

The recommended operating conditions are list in **Table 4-2**.

**Table 4-2** Recommended operating conditions

| Symbol | Parameter | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| VDDCORE | Core supply voltage | 0.81 | 0.9 | 0.99 | V |
| VDDARM | Arm® core supply voltage | 0.99 | 1.1 | 1.155 | V |
| AVDD1V8_BB | Baseband supply voltage | 1.7 | 1.8 | 1.98 | V |
| AVDD1V8_MDPLL | Analog PLL supply voltage | 1.7 | 1.8 | 1.98 | V |
| AVDD1V8_DSI | MIPI DSI supply voltage | 1.7 | 1.8 | 1.98 | V |

| Symbol | Parameter | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| AVDD1V8_CSI | MIPI CSI supply voltage | 1.7 | 1.8 | 1.98 | V |
| AVDD1V8_LVDSRF | LVDSRF supply voltage | 1.7 | 1.8 | 1.98 | V |
| AVDD1V8_USB | USB PHY supply voltage | 1.7 | 1.8 | 1.98 | V |
| AVDD3V3_USB | USB PHY supply voltage | 3.0 | 3.3 | 3.6 | V |

## 4.1.3 Thermal characteristics

The thermal characteristics are as shown in **Table 4-3**.

**Table 4-3** Thermal characteristics

| Symbol | Parameter | Condition | Value | Unit |
|---|---|---|---|---|
| Theta JA | Junction-to-Ambient thermal resistance | Air flow: 0 m/s | 23.98 | °C/W |

## 4.1.4 ESD characteristics

The ESD characteristics are shown in **Table 4-4**.

**Table 4-4** ESD characteristics

| Symbol | Parameter | Condition | Value | Unit |
|---|---|---|---|---|
| HBM | Human body model | MIL STD 883G, method 3015.7 | ±2K | V |
| CDM | Charged-device model | JESD22-C101-C | ±500 | V |
| LU | Latch UP current | JESD78E | ±100 | mA |

## 4.1.5 DC characteristics

The typical core voltage (VDDCORE) is 0.9V and the digital I/O supply (VIO1V8) is typically at 1.85V. The power pins should be connected with a decoupling capacitor to ground (VSS).
For the following table, VSS = 0V (ground), and all voltages are measured with respect to VSS, unless otherwise specified.

**Table 4-5** DC characteristics

| Symbol | Parameter | Conditions | Min | Typical | Max | Unit |
|---|---|---|---|---|---|---|
| **Digital supply voltage: pins VIO1V8** | | | | | | |
| VIO1V8 | Digital supply voltage | | 1.62 | – | 1.98 | V |
| **Digital supply voltage: pins VIOCTP** | | | | | | |
| VIOCTP | Digital supply voltage | | 1.62 | – | 3.3 | V |
| **Digital supply voltage: pins VSIM0** | | | | | | |

| Symbol | Parameter | Conditions | Min | Typical | Max | Unit |
|---|---|---|---|---|---|---|
| VSIM0 | Digital supply voltage | | 1.62 | – | 3.3 | V |
| **Digital supply voltage: pins VSIM1** | | | | | | |
| VSIM1 | Digital supply voltage | | 1.62 | – | 3.3 | V |
| **Digital supply voltage: pins VSIM2** | | | | | | |
| VSIM2 | Digital supply voltage | | 1.62 | – | 3.3 | V |
| **Digital supply voltage: pins VSD0** | | | | | | |
| VSD0 | Digital supply voltage | | 1.62 | – | 3.3 | V |
| **Digital supply voltage: pins VCAM** | | | | | | |
| VCAM | Digital supply voltage | | 1.62 | – | 1.98 | V |
| **Digital supply voltage: pins VDDMEM** | | | | | | |
| VDDMEM | Digital supply voltage | | 1.1 | – | 1.98 | V |
| **Digital supply voltage: pins VSD2** | | | | | | |
| VSD2 | Digital supply voltage | | 1.62 | – | 3.3 | V |
| **Digital supply voltage: pins VEMMCIO** | | | | | | |
| VEMMCIO | Digital supply voltage | | 1.62 | – | 1.98 | V |
| **Digital supply voltage: internal VDDCORE** | | | | | | |
| VDDCORE | Digital core supply voltage | | 0.81 | 0.9 | 0.99 | V |
| | Sleep mode digital core supply voltage | | - | 0.7 | - | V |
| | Arm® Core supply voltage | | 0.99 | 1.1 | 1.155 | V |
| **Digital input** | | | | | | |
| $V_{IL}$ | Input voltage Low-level | | 0 | – | $0.3V_{pad}$ | V |
| $V_{IH}$ | Input voltage High-level | | $0.7V_{pad}$ | – | $V_{pad}$ | V |
| $I_{LI}$ | Input leakage current | | – | ±2 | – | μA |

| Symbol | Parameter | Conditions | Min | Typical | Max | Unit |
|---|---|---|---|---|---|---|
| **Digital output** | | | | | | |
| $V_{OL}$ | Output voltage Low-level | At $I_{sink}$=2,4,6,8mA (programmable) | 0 | – | $0.1V_{pad}$ | V |
| $V_{OH}$ | Output voltage High-level | At $I_{source}$=2,4,6,8 mA(programmable) | $0.9V_{pad}$ | – | $V_{pad}$ | V |
| **Analog supply voltage** | | | | | | |
| AVDD1V8_BB | Baseband supply voltage | | 1.7 | 1.8 | 1.98 | V |
| AVDD1V8_MDPLL | Analog pll supply voltage | | 1.7 | 1.8 | 1.98 | V |
| AVDD1V8_DSI | MIPI DSI supply voltage | | 1.7 | 1.8 | 1.98 | V |
| AVDD1V8_CSI | MIPI CSI supply voltage | | 1.7 | 1.8 | 1.98 | V |
| AVDD1V8_LVDSRF | LVDSRF supply voltage | | 1.7 | 1.8 | 1.98 | V |
| AVDD1V8_USB | USB PHY supply voltage | | 1.7 | 1.8 | 1.98 | V |
| AVDD3V3_USB | USB PHY supply voltage | | 3.0 | 3.3 | 3.6 | V |

$V_{pad}$ means the power supply voltage at the corresponding pad.

## 4.2 AC characteristics

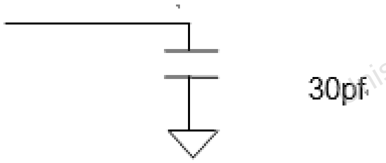
A pin's AC characteristics include input and output capacitance, which determine loading for external drivers or other load analysis. The AC characteristics also include a de-rating factor, which indicates how much faster or slower the AC timings get with different loads.

**Table 4-6** Standard input, output and I/O pin AC characteristics

| Symbol | Parameters | Min | Typical | Max | Units |
|---|---|---|---|---|---|
| $C_{in}$ | Input capacitance, all standard input and IO pins | | | 3.5 | pF |
| $C_{load}$ | Output capacitance, all standard output and IO pins | | | 30 | pF |
| $T_{dr}$ | Output de-rating falling edge on all standard output and I/O pins, from 30 pF load | | 0.075 | | ns/pF |

The AC specifications are tested with a 30 pF load as indicated in **Figure 4-1**

**Figure 4-1** Test circuit of an I/O pin




The output capacitance and de-rating falling edge are measured under the condition of maximum driving strength: 24 mA@1.8V.

For the following tables, VSS = 0V (ground), and all voltages are measured with respect to VSS, unless otherwise specified.

**Table 4-7** System Clock AC characteristics

| Symbol | Parameter | Conditions | Min | Typical | Max | Unit |
|---|---|---|---|---|---|---|
| $f_{mclk}$ | Master clock frequency | | – | 26 | – | MHz |
| $C_I$ | Input capacitance | | – | 30 | – | pF |
| **Master clock input** | | | | | | |
| Vpp | Sine wave peak-to-peak swing | Loading = 30pF | 0.5 | 0.6 | 1.8 | V |
| **Real-time clock input** | | | | | | |
| $f_{rtclk}$ | Real time clock frequency | | – | 32 | – | kHz |
| $\triangle f/f_{rtclk}$ | Frequency tolerance | Standard deviation | -80 | | 80 | ppm |
| $\delta_{rtclk}$ | Clock duty cycle | | 35 | – | 65 | % |

# 4.3 Performance specification

## 4.3.1 LVDSRF

The LVDS interface is designed for transmitting and receiving data between RF chip and BB chip. The recommended line rate is 1.04Gbps.

### 4.3.1.1 DC specifications

**Table 4-8** LVDS TX DC specifications

| Symbol | Descriptions | Specifications | | | Unit |
|---|---|---|---|---|---|
| | | Min | Typical | Max | |
| $\|V_{od}\|$ | Transmit Differential output voltage magnitude | 190 | 200 | 210 | mV |
| $\triangle\|V_{od}\|$ | Change in Differential output voltage magnitude between logic states | | | 25 | mV |
| $V_{cmtx}$ | Steady-state common-mode output voltage | 180 | 200 | 220 | mV |

| Symbol | Descriptions | Specifications | | | Unit |
|---|---|---|---|---|---|
| | | Min | Typical | Max | |
| $\Delta V_{cmtx}$ | Changes in steady-state common-mode output voltage between logic states | | | 25 | mV |
| $Z_{os}$ | Single-ended output impedance | | 50 | | Ω |

**Table 4-9** LVDS RX DC specifications

| Symbol | Descriptions | Specifications | | | Unit |
|---|---|---|---|---|---|
| | | Min | Typical | Max | |
| $V_{IDTH}$ | Differential input high voltage threshold | | | 50 | mV |
| $V_{IDTL}$ | Differential input low voltage threshold | -50 | | | mV |
| $V_{IHHS}$ | Single ended input high voltage | | | 1100 | mV |
| $V_{ILHS}$ | Single ended input low voltage | -40 | | | mV |
| $V_{CMRXDC}$ | Input common mode voltage | 50 | 200 | 950 | mV |
| $Z_{ID}$ | Differential input impedance (before calibration) | 80 | 100 | 125 | Ω |

### 4.3.1.2 AC specifications

**Table 4-10** LVDS TX AC specifications

| Symbol | Descriptions | Specifications | | | Unit |
|---|---|---|---|---|---|
| | | Min | Typical | Max | |
| $C_{near}$ | Near-end capacitor to ground | | 2 | 3 | pF |
| $C_{far}$ | Far-end capacitor to ground | | 2 | 3 | pF |
| $t_r$ | Differential output signal rise time (20%-80%) | 90 | | 270 | ps |
| $t_f$ | Differential output signal fall time (80%-20%) | 90 | | 270 | ps |
| $\Delta V_{cmtx}$ | Changes in common-mode output voltage when random bit stream in | | 50 | 100 | mVpp |

**Table 4-11** LVDS RX AC specifications

| Symbol | Descriptions | Specifications | | | Unit |
|---|---|---|---|---|---|
| | | Min | Typical | Max | |
| $\Delta V_{CMRX(HF)}$ | Common mode interference beyond 450 MHz | | | 200 | mVpp |

| Symbol | Descriptions | Specifications | | | Unit |
|---|---|---|---|---|---|
| | | Min | Typical | Max | |
| $\Delta V_{CMRX(LF)}$ | Common mode interference between 50MHz and 450MHz | -50 | | 50 | mVpp |
| $C_{CM}$ | Common mode termination | | | 10 | pF |

**Table 4-12** Operating Voltage and Temperature Specification

| Description | Min | Typical | Max | Unit |
|---|---|---|---|---|
| Analog Supply Voltage | 1.7 | 1.8 | 1.98 | V |
| Digital Core Supply Voltage | 0.81 | 0.9 | 0.99 | V |

## 4.3.2 SOC Phase-Locked Loop (PLL)

**Table 4-13** Performance of the phase-locked loop

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Input reference | | | 26 | | MHz |
| Frequency Range | MPLL | 800 | | 1600 | MHz |
| | RPLL | 780 | | 1600 | MHz |
| | CPPLL | 450 | | 1000 | MHz |
| | LPLL | 800 | 1228.8 | 1600 | MHz |
| | GPLL | 400 | | 800 | MHz |
| | TWPLL | 800 | 1536 | 1600 | MHz |
| Long term jitter of LVDSPLL | LVDSPLL | | | 50 | ps |
| Settling time | | | | 100 | µs |
| Sleep mode | Yes | | | | |

## 4.3.3 Automatic Power Control (APC) DAC

One general DAC is used to control power ramping and gain. It is a 10 bits D/A converters with a programmable-gain output driver. Special offset cancellation is applied in the DAC.

**Table 4-14** Performance of the APC DAC

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Power supply | Analog supply voltage(AVDD)<br>Digital core supply voltage(DVDD) | 1.7<br>0.81 | 1.8<br>0.9 | 1.98<br>0.99 | V<br>V |
| Max Output range | PGA gain=1.3 | 1.34*(1-4.5%) | 1.34 | 1.34*(1+4.5%) | V |
| | PGA gain=1.5 | 1.508*(1-4.5%) | 1.508 | 1.508*(1+4.5%) | V |
| | PGA gain=2.0 | | AVDD-50mV | | V |
| N | Resolution of the DAC | | 14 | | bit |
| Sample rate | | | | 1 | MHz |
| PGA gain | | 1.3 | 1.5 | 2 | |
| DNL | Differential nonlinearity | | ±1 | ±2 | LSB |
| Setting time | 4pF Capacitor load, full scale input, 1.5X PGA gain, bypass internal RC filter. | | | 1 | µs |
| PSRR | | 45 | | | dB |

### 4.3.4 CLK26M Buffer

This CLK26M Buffer is used to generate 26MHz clock for PLL and other circuit blocks. It includes one 26M_buffer and one sine driver. Output of this CLK26M_TOP block can be set the sine wave or the square wave.

**Table 4-15** Performance of the 26MHz Buffer

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Power supply | Analog supply voltage<br>Digital core supply voltage | 1.7<br>0.81 | 1.8<br>0.9 | 1.98<br>0.99 | V<br>V |
| $Vpp_{\_in}$ | Input sine wave peak to peak amplitude | 500 | 600 | 1800 | mV |
| $f_{\_26}$ | Output Clock Frequency | | 26 | | MHz |
| Duty cycle | Output Clock Duty cycle | 45 | 50 | 55 | % |
| $f_{\_sine}$ | Output sine wave Frequency | | 26 | | MHz |
| $Vpp_{\_o}$ | Output sine wave peak-to-peak amplitude | | 500 | | mV |

## 4.3.5 Thermal Sensor

This is a temperature-to-digital converter using an on-chip bandgap temperature sensor. The overall accuracy is ±3°C (typical) from 0°C to 90°C.

This is suitable for multi-core smart phone applications requiring highly accurate temperature sensing to perform thermal control.

**Table 4-16** Performance of the Thermal sensor

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Power supply | Analog supply voltage<br>Digital core supply voltage | 1.7<br>0.81 | 1.8<br>0.9 | 1.98<br>0.99 | V<br>V |
| Temperature resolution | | | 0.9 | | °C |
| Temperature accuracy | 0°C ~90°C | | ±3 | | °C |
| Temperature accuracy | -20°C ~0°C, 90°C ~ 125°C | | ±5 | | °C |
| Conversion time | | | 29 | | Clock cycle |

## 4.3.6 USB 2.0 PHY

The USB 2.0 PHY in this chip is fully compliant with USB 2.0 specifications. Its own PLL can generate 480MHz clock for its usage. Please refer to Universal Serial Bus Specification revision 2.0 at http://www.usb.org/developers/docs

**Table 4-17** Main performance of the USB

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Power supply | Analog 1.8V power supply<br>Analog 3.3V power supply<br>Digital core supply voltage | 1.7<br>3.0<br>0.81 | 1.8<br>3.3<br>0.9 | 1.98<br>3.6<br>0.99 | V |
| $T_{HSR}$ | Rise time(10%~90% ) | 500 | | | ps |
| $T_{HSF}$ | Fall time (90%~10%) | 500 | | | ps |
| $Z_{out\ DRV(HS)}$ | Driver output impedance | 40.5 | 45 | 49.5 | Ω |
| $V_{CRS(FS)}$ | Crossover voltage | 1.3 | | 2.0 | V |
| $TFRFM_{(FS)}$ | Rise time and fall time matching | 90 | 100 | 110 | % |
| $T_{FR(FS)}$ | Rise time | 4 | | 20 | ns |
| $T_{FF(FS)}$ | Fall time | 4 | | 20 | ns |
| $R_{PU1}$ | Bus pull up resistor1 on upstream port | 900 | | 1575 | Ω |
| $R_{PU2}$ | Bus pull up resistor2 on upstream port | 525 | | 1515 | Ω |
| $R_{PD}$ | Bus pull down resistor on downstream port | 14.25k | | 24.8k | Ω |
| $V_{DI}$ | Differential input sensitivity | 0.2 | | | V |
| $V_{CM}$ | Differential input common mode range | 0.8 | | 2.5 | V |

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| $T_{FST(FS)}$ | Width of SE0 during differential transition | | | 14 | ns |
| $Z_{out\ DRV(FS)}$ | Driver output resistor | 28 | | 44 | Ω |
| $V_{CRS(LS)}$ | Crossover voltage | 1.3 | | 2.0 | V |
| $TFRFM_{(LS)}$ | Rise time and fall time matching | 80 | 100 | 120 | % |
| $T_{FR(LS)}$ | Rise time | 75 | | 200 | ns |
| $T_{FF(LS)}$ | Fall time | 75 | | 200 | ns |
| $V_{HSSQ}$ common range | $V_{HSSQ}$ common input range | -50 | | 500 | mV |
| $T_{FST(LS)}$ | Width of SE0 during differential transition | | | 210 | ns |

## 4.3.7 MIPI CSI

MIPI CSI is fully compliant with MIPI CSI-2 V1.2 and D-PHY 1.2. Please refer specification at http://www.mipi.org

**Table 4-18** Main performance of the MIPI CSI

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Power supply | Analog supply voltage<br>Digital core supply voltage | 1.7<br>0.81 | 1.8<br>0.9 | 1.98<br>0.99 | V |
| **HS RX mode** | | | | | |
| $V_{cmrx}$ | Common-mode voltage HS receiver mode | 70 | | 330 | mV |
| $V_{idth}$ | Differential input high threshold | | | 70 | mV |
| $V_{idtl}$ | Differential input low threshold | -70 | | | mV |
| $V_{ihhs}$ | Single-ended input high voltage | | | 460 | mV |
| $V_{ilhs}$ | Single-ended input low voltage | -40 | | | mV |
| $Z_{id}$ | Differential input impedance | 80 | | 125 | Ω |
| $\triangle V_{cmrx(HF)}$ | Common-mode interference beyond 450MHz | | | 100 | mV |
| $\triangle V_{cmrx(LF)}$ | Common-mode interference beyond 50-450MHz | -50 | | 50 | mV |
| **LP RX mode** | | | | | |
| $V_{IH}$ | Logic high | 880 | | | mV |
| $V_{IL}$ | Logic low | | | 550 | mV |
| $V_{IL-ULPS}$ | Logic low@ULP | | | 300 | mV |
| $V_{HYST}$ | Input hysteresis | 25 | | | mV |
| $e_{SPIKE}$ | Input pulse rejection | | | 300 | V•ps |
| $T_{MIN-RX}$ | Minimum pulse width response | 50 | | | ns |

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| **Data-Clock Timing Spec** | | | | | |
| $T_{setup(rx)}$@1.5Gbps | | 0.2 | | | UI |
| $T_{setup(rx)}$@1Gbps | | 0.15 | | | UI |
| $T_{hold(rx)}$@1.5Gbps | | 0.2 | | | UI |
| $T_{hold(rx)}$@1Gbps | | 0.15 | | | UI |

## 4.3.8 MIPI DSI

MIPI DSI is fully compliant with MIPI DSI V1.2 and D-PHY 1.2. Please refer specification at http://www.mipi.org

**Table 4-19** Main performance of the MIPI DSI

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Power supply | Analog supply voltage<br>Digital core supply voltage | 1.7<br>0.81 | 1.8<br>0.9 | 1.98<br>0.99 | V |
| **HS TX mode** | | | | | |
| $Z_{os}$ | Single ended output impedance | 40 | 50 | 62.5 | Ω |
| $V_{CMTX}$ | HS transmit static common-mode voltage | 150 | 200 | 250 | mV |
| $\|\triangle V_{CMTX(1,0)}\|$ | $V_{CMTX}$ mismatch when output is Differential-1 or 0 | | | 5 | mV |
| $\|V_{OD}\|$ | HS transmit differential voltage | 140 | 200 | 270 | mV |
| $\|\triangle V_{OD}\|$ | $V_{OD}$ mismatch when output is Differential-1 or 0 | | | 10 | mV |
| $V_{OHHS}$ | HS output high voltage | | | 360 | mV |
| $\triangle V_{CMTX(HF)}$ | Common-level variations above 450MHz | | | 15 | mV RMS |
| $\triangle V_{CMTX(LF)}$ | Common-level variation between 50~450MHz | | | 25 | mV |
| $t_r$ / $t_f$ | 20%~80% rise time and fall time(<1.5Gbps) | 100 | | 0.35UI | ps |
| **LPTX mode** | | | | | |
| $V_{OH}$ | Thevenin output high level | 1.1 | 1.2 | 1.3 | V |
| $V_{OL}$ | Thevenin output low level | -50 | | 50 | mV |
| $Z_{OLP}$ | Output impedance of LP transmitter | 110 | | | Ω |
| ΔZOLP(01,10) | Single-ended output impedance mismatch driving opposite level | | | 20 | % |
| ΔZOLP(00,11) | Single-ended output impedance mismatch driving same level | | | 5 | % |
| Slew rate | $C_{load}$=70p | 30 | | 120 | mV/ns |

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| **LPRX mode** | | | | | |
| $V_{IH}$ | Logic high | 880 | | | mV |
| $V_{IL}$ | Logic low | | | 550 | mV |
| $V_{HYST}$ | Input hysteresis | 25 | | | mV |
| **LPCD mode** | | | | | |
| $V_{IHCD}$ | Logic high contention threshold | 450 | | | mV |
| $V_{ILCD}$ | Logic low contention threshold | | | 200 | mV |

## 4.3.9 Temperature sensing ADC

**Table 4-20** Main performance of the temperature sensing ADC

| Parameter | Conditions/Comments | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| $V_{REFP}$ | Reference Voltage | 1.45 | 1.5 | 1.6 | V |
| Resolution | Power supply 1.5V | | 16 | | bit |
| Integral non-linearity | Power supply 1.5V | -2 | | +2 | LSB |
| Maximum Analog input voltage | Power supply 1.5V | | 0.91*VREFP | | V |
| Minimum Analog input voltage | Power supply 1.5V | | 0.091*VREFP | | V |
| Clock frequency | Power supply 1.5V | | 6.5 | 13 | MHz |
| Conversion time | Power supply 1.5V | 5.04 | 10.08 | | ms |

## 4.3.10 Wi-Fi RF

The specification value is valid at room temperature (25℃).

Unless otherwise specified, all specifications are measured at the RF antenna port.

### 4.3.10.1 Wi-Fi Receiver Specifications

**Table 4-21** Wi-Fi Receiver Specifications

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Frequency range | | | - | | MHz |
| RX sensitivity | 1 Mbps DSSS | | -93.5 | | dBm |
| | 11 Mbps DSSS | | -84.5 | | dBm |
| RX Sensitivity | 6 Mbps OFDM | | -88.5 | | dBm |

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| | 54 Mbps OFDM | | -72 | | dBm |
| RX sensitivity BW=20MHz Green field 800ns guard interval Non-STBC | MCS 0 | | -89 | | dBm |
| | MCS 7 | | -69 | | dBm |
| Maximum input level | 11 Mbps DSSS | | 0 | | dBm |
| | 6 Mbps OFDM | | 0 | | dBm |
| | 54 Mbps OFDM | | -3 | | dBm |
| | MCS0 | | 0 | | dBm |
| | MCS7 | | -3 | | dBm |
| Adjacent channel rejection (30MHz offset) | 1 Mbps DSSS | | 5 | | dB |
| Adjacent channel rejection (25MHz offset) | 11 Mbps DSSS | | 4 | | dB |
| Adjacent channel rejection (25MHz offset) | 6 Mbps OFDM | | 9 | | dB |
| | 54 Mbps OFDM | | 8 | | dB |
| Adjacent channel rejection (25MHz offset), BW=20MHz | MCS 0 | | 13 | | dB |
| | MCS 7 | | 12 | | dB |

#### 4.3.10.2 Wi-Fi Transmitter Specifications

**Table 4-22** Wi-Fi Transmitter Specifications

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Frequency range | | | - | | MHz |
| Output power | 802.11b, 1~11 Mbps DSSS | | 16 | | dBm |
| | 802.11g, 6 ~54Mbps OFDM | | 14 | | dBm |
| | 802.11n, HT20 MCS0~7 | | 13 | | dBm |
| EVM | 802.11b, 1~11 Mbps DSSS @Pout=17dBm | | 22 | | % |
| | 802.11g, 6 ~54Mbps OFDM @Pout=14dBm | | -29.5 | | dB |
| | 802.11n, HT20 MCS0~7 @Pout=14dBm | | -29.5 | | dB |

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| TX power accuracy | -40~85 °C, 2~18dBm | | 1.8 | | dB |
| Transmitted power (Data rate =1M, Pout=20dBm) | 76 ~ 108 MHz | | -111.7 | | dBm/Hz |
| | 776 ~ 794 MHz | | -111.4 | | dBm/Hz |
| | 869 ~ 960 MHz | | -111.1 | | dBm/Hz |
| | 1,570 ~ 1,580 MHz | | -111.6 | | dBm/Hz |
| | 1,805 ~ 1,880 MHz | | -110.4 | | dBm/Hz |
| | 1,930 ~ 1,990 MHz | | -110.4 | | dBm/Hz |
| | 2,110 ~ 2,170MHz | | -109.7 | | dBm/Hz |
| Harmonic output power (Data rate =1M, Pout=20dBm) | 2nd harmonic | | -108.7 | | dBm/Hz |
| | 3rd harmonic | | -108.9 | | dBm/Hz |

## 4.3.11 Bluetooth RF

The specification value is valid at room temperature (25°C).

Unless otherwise specified, all specifications are measured at the RF antenna port.

### 4.3.11.1 Bluetooth BDR Specifications

**Table 4-23** Bluetooth BDR receiver specifications

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Frequency range | | | - | | MHz |
| RX sensitivity | BER < 0.1% | | -90 | | dBm |
| Max. input level | BER < 0.1% | | -4 | | dBm |
| C/I co-channel | Co-channel selectivity (BER<0.1%) | | 5 | | dB |
| C/I 1MHz | Adjacent channel selectivity (BER < 0.1%) | | -16 | | dB |
| C/I 2MHz | 2nd adjacent channel selectivity (BER < 0.1%) | | -40 | | dB |
| C/I ≧3MHz | 3rd adjacent channel selectivity (BER < 0.1%) | | -43 | | dB |
| C/I image channel | Image channel selectivity (BER<0.1%) | | -26 | | dB |
| C/I image 1MHz | 1MHz adjacent to image channel selectivity (BER<0.1%) | | -43 | | dB |

**Table 4-24** Bluetooth BDR transmitter specifications

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Frequency range | | | - | | MHz |
| Output power | At max power output level | | 8 | | dBm |
| Modulation characteristic | $\triangle f1_{avg}$ | | 160 | | kHz |
| | $\triangle f2_{max}$ (for at least 99% of all $\triangle f2_{max}$) | | 115 | | kHz |
| | $\triangle f2_{avg}/\triangle f1_{avg}$ | | 0.92 | | |

### 4.3.11.2 Bluetooth EDR Specifications

**Table 4-25** Bluetooth EDR Receiver Specifications

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Frequency range | | | - | | MHz |
| Receiver sensitivity | π/4 DQPSK (BER<0.01%) | | -90 | | dBm |
| | 8PSK (BER<0.01%) | | -84 | | dBm |
| Max. input level | π/4 DQPSK (BER<0.1%) | | -4 | | dBm |
| | 8PSK (BER<0.1%) | | -4 | | dBm |
| C/I co-channel | π/4 DQPSK (BER<0.1%) | | 9 | | dB |
| | 8PSK (BER<0.1%) | | 15 | | dB |
| C/I 1MHz | π/4 DQPSK (BER<0.1%) | | -14 | | dB |
| | 8PSK (BER<0.1%) | | -8 | | dB |
| C/I 2MHz | π/4 DQPSK (BER<0.1%) | | -42 | | dB |
| | 8PSK (BER<0.1%) | | -36 | | dB |
| C/I ≧3MHz | π/4 DQPSK (BER<0.1%) | | -43 | | dB |
| | 8PSK (BER<0.1%) | | -37 | | dB |
| C/I image channel | π/4 DQPSK (BER<0.1%) | | -26 | | dB |
| | 8PSK (BER<0.1%) | | -22 | | dB |
| C/I image 1MHz | π/4 DQPSK (BER<0.1%) | | -43 | | dB |
| | 8PSK (BER<0.1%) | | -35 | | dB |

**Table 4-26** Bluetooth EDR Transmitter Specifications

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Frequency range | | | - | | MHz |
| Output power | π/4 DQPSK | | 6 | | dBm |

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| | 8PSK | | 6 | | dBm |
| Modulation accuracy | π/4 DQPSK(RMS DEVM) | | 6.5 | | % |
| | 8PSK(RMS DEVM) | | 6.5 | | % |
| | π/4 DQPSK(99% DEVM) | | 8 | | % |
| | 8PSK(99% DEVM) | | 8 | | % |
| | π/4 DQPSK(Peak DEVM) | | 13 | | % |
| | 8PSK(Peak DEVM) | | 13 | | % |
| In-band spurious emission | π/4 DQPSK(±1MHz offset) | | -30 | | dB |
| | 8PSK(±1MHz offset) | | -30 | | dB |
| | π/4 DQPSK(±2MHz offset) | | -32 | | dBm |
| | 8PSK(±2MHz offset) | | -31 | | dBm |
| | π/4 DQPSK(±3MHz offset) | | -42 | | dBm |
| | 8PSK(±3MHz offset) | | -42 | | dBm |

### 4.3.11.3 Bluetooth LE Receiver Specifications

**Table 4-27** Bluetooth LE Receiver Specifications

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Frequency range | | | - | | MHz |
| Receiver sensitivity | PER<30.8% | | -95 | | dBm |
| Max. input signal level | PER<30.8% | | -4 | | dBm |

**Table 4-28** Bluetooth LE Transmitter Specifications

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Frequency range | | | - | | MHz |
| Output power | At max. power output level | | 0 | | dBm |
| Modulation characteristic | $\triangle f1_{avg}$ | | 247 | | kHz |
| | $\triangle f2_{max}$ (For at least 99% of all $\triangle f2_{max}$) | | 185 | | kHz |
| | $\triangle f2_{avg}/\triangle f1_{avg}$ | | 0.94 | | |

## 4.3.12 FM RF

Typical specifications are for channel 98MHz, default register settings and under recommended operating conditions. The min/max specifications are for extreme operating voltage and temperature conditions, unless otherwise stated.

**Table 4-29** FM RF Specifications

| Parameter | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|
| Input frequency range | | | | | MHz |
| Sensitivity1,3 | (S+N)/N=26dB, unmatched | | 5 | | dBuVemf |
| | (S+N)/N=26dB, matched | | 3 | | dBuVemf |
| RDS sensitivity | △f=2kHz, BLER<5%, unmatched | | 18 | | dBuVemf |
| Audio mono (S+N)/N1,3,4 | | | 60 | | dB |
| Audio stereo (S+N)/N2,3,4 | | | 60 | | dB |
| Audio stereo separation4 | f=75kHz | | 50 | | dB |
| Audio output THD1,4 | | | 0.3 | | % |
| 1 D△f=22.5kHz, fm=1kHz, 75μs de-emphasis, mono, L=R | | | | | |
| 2 D△f=22.5kHz, fm =1kHz, 75μs de-emphasis, stereo | | | | | |
| 3 A-weighting, BW=300Hz to 15kHz | | | | | |
| 4 Vin = 60dBuVemf | | | | | |

## 4.3.13 GNSS RF

**Table 4-30** GNSS RF Specifications

| Parameter | | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|---|
| Frequency range | | | | 1575.42 | | MHz |
| sGPS_C/No_Conducted | | Power Levels All SV=-130dBm | | 40 | | dB*Hz |
| sGPS_Acquisition Sensitivity_cold | | Time out for first fix=300s | | -148.5 | | dBm |
| sGPS_Tracking Sensitivity | | | | -165 | | dBm |
| sGPS_Simulator_static_cold_TTFF@-130dBm | ave | Number of Test=30 | | 26 | | s |
| sGPS_Simulator_static_cold_Accuracy@-130dBm | ave | Number of Test=30 | | 0.8 | | m |

| Parameter | | Description | Min | Typical | Max | Unit |
|---|---|---|---|---|---|---|
| sGPS_Simulator_static_cold_TTFF@-140dBm | ave | Number of Test=30 | | 31 | | s |
| sGPS_Simulator_static_cold_Accuracy@-140dBm | ave | Number of Test=30 | | 0.8 | | m |